복사전달 특강

Special Topics in Radiative Transfer

Lecture 5

2026 March 31 (Tuesday), 1PM

updated on 03/31

선광일 (Kwangil Seon)
KASI / UST

[Zeeman Effect]

- **Zeeman effect**

  the splitting of atomic energy levels in the presence of an external magnetic field.

- **Zeeman effect at 1420MHz (21.1 cm)**

  (1) The upper level of the 1420 MHz hyperfine transition of hydrogen is the $F = 1$ state. Its angular momentum has three possible projections onto a given axis. The three states are

$$F_z = m_\mathrm{F}\hbar \quad (m_\mathrm{F} = -1, 0, +1)$$

The three states have the same energy (i.e., they are degenerate).

However, if the atoms are immersed in an external magnetic field, the three states take on different energies. This is a consequence of the interaction of the magnetic moment of the hydrogen atom with the external field.

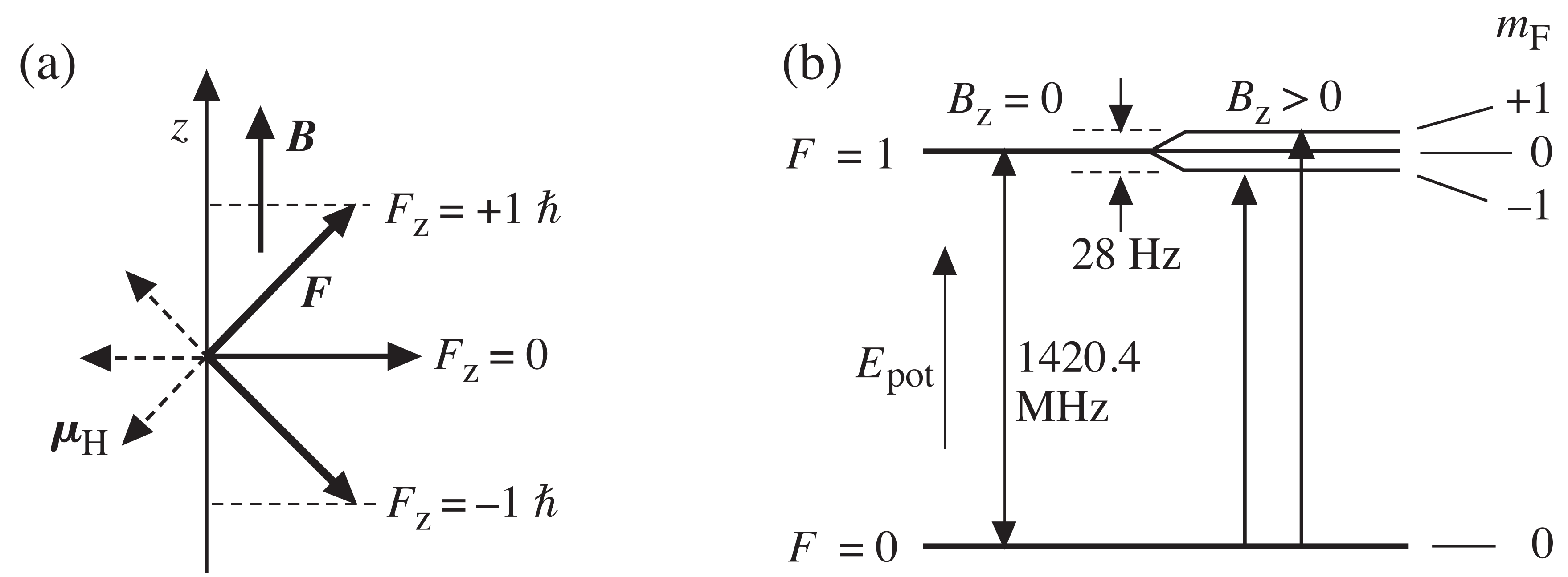
$$\boxed{\begin{aligned} E_\mathrm{pot} &= -\boldsymbol{\mu}\cdot\mathbf{B} \simeq g_e \frac{e}{2m_e}\mathbf{F}\cdot\mathbf{B} \\ &= g_e \frac{e\hbar}{2m_e} m_\mathrm{F} B \end{aligned}}$$

$\mu \simeq \mu_e$ because μ arises almost totally from the electron.

The separation between the +1 and -1 levels is $\boxed{\Delta E_\mathrm{pot} = E_\mathrm{pot}(m_\mathrm{F} = +1) - E_\mathrm{pot}(m_\mathrm{F} = -1) = g_e \frac{e\hbar}{m_e} B}$

(2) In the $F = 0$ ground state, the total angular momentum is zero. Therefore, the magnetic moment is zero and no Zeeman splitting occurs in the ground state.

Photons carry one unit of angular momentum $\mathbf{J}$. The allowed transitions are from the $F = 0$ $(m_F = 0)$ to the $F = 1$ $(m_F = +1)$ or $F = 1$ $(m_F = -1)$.



Astrophysics Processes [Bradt]

Energy levels for Zeeman absorption by hydrogen atom in an external magnetic field $\mathbf{B}$. The z-axis is taken to be in the direction of $\mathbf{B}$.

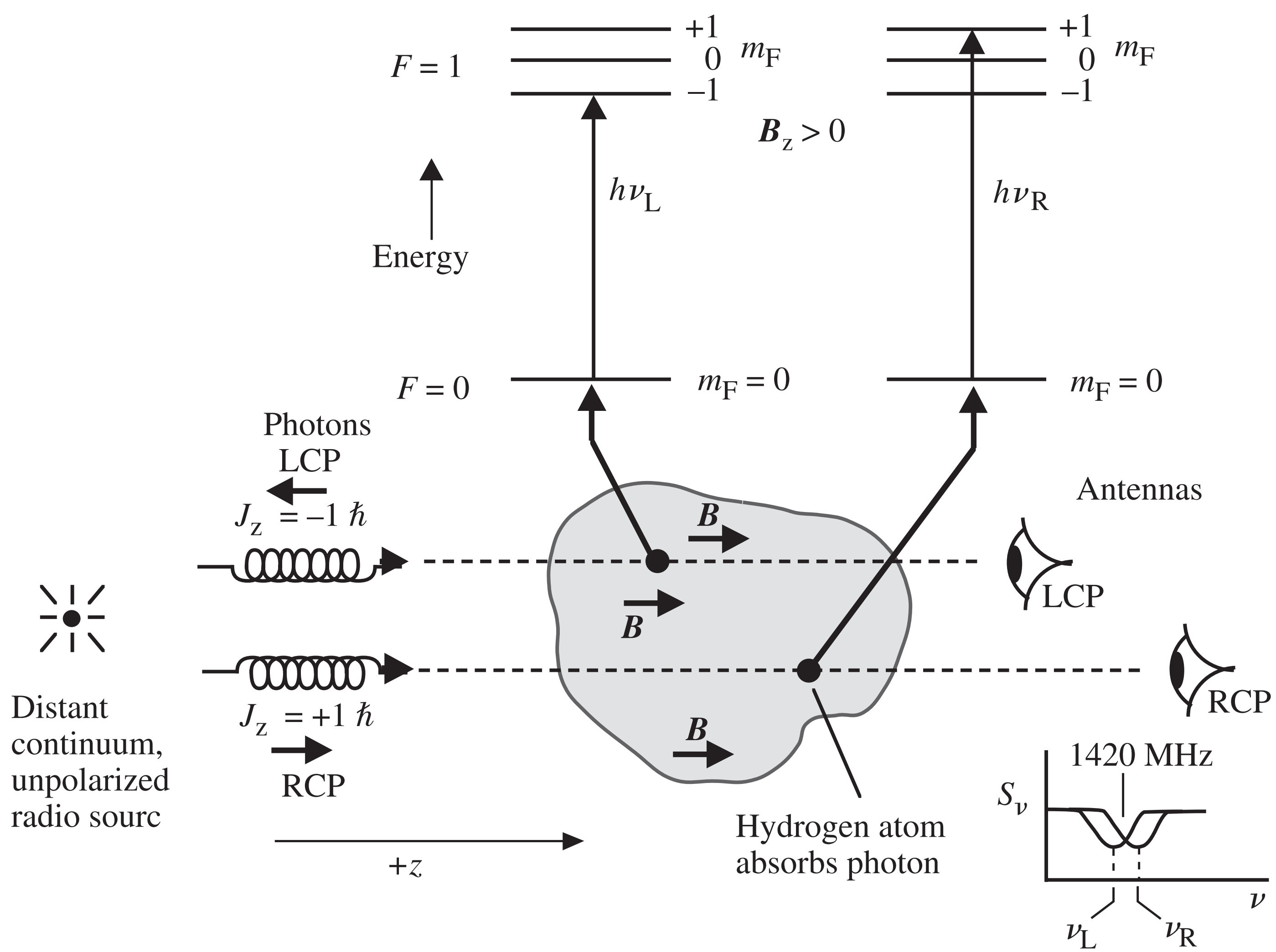
(a) The three allowed projections $(m_F = +1, 0, -1)$ of the angular momentum for the $F = 1$ state. The atomic magnetic moment $\boldsymbol{\mu}_H$, which is directed opposite to the angular momentum, interacts with the magnetic field to perturb the energy state.

(b) Energy levels showing (left) the 1420-MHz transition for $B = 0$ and (right) the Zeeman splitting of the upper energy level into the three sublevels for $B_z > 0$. The two allowed transitions to the upper states are shown; they differ in frequency by 28 Hz for $B_z = 1$ nT (= 10μG).

The typical magnetic field strength in the ISM is ~ 3–6 μG near the Sun, and is higher in dense clouds and near the Galactic center (~10–100 μG).

In quantum theory, a single photon is always circularly polarized with $S = \pm\hbar$ (no state with $S = 0$).
Linearly polarized photon state is a superposition of a pair of circularly polarized photons.

A magnetic field of order 10 μG shifts the frequency by about one part in 10^8 of the hyperfine splitting. This shift is much smaller than the frequency shift $v/c \sim 10^{-5}$ due to a radial velocity of a few km s^{-1}, and it would be nearly impossible to detect, except that it leads to a shift in frequency between two circular polarization modes. The Zeeman effect in H I 21 cm can therefore be detected by taking the difference of the two circular polarization signals. This technique has been used to measure the magnetic field strength in a number of H I regions.



$$\nu_R - \nu_L = \frac{eB_z}{2\pi m_e} = 28.0 \left(\frac{B_z}{1.0\,\mathrm{nT}}\right) \mathrm{Hz}$$

Left circularly polarized (LCP) photons are absorbed at a slightly lower frequency.
Right circularly polarized (RCP) photons are absorbed at a slightly higher frequency.
(Bradt: Astrophysics Processes)

See also Heiles & Robishaw (2009)

Spectral Line Broadening

- Natural broadening (radiation damping): due to the limited lifetimes of excited states
- Doppler broadening: by thermal motions
- Collisional broadening: due to collisions with or perturbation by other particles

Natural Broadening

- The natural line broadening profile described by the Lorentz profile:

$$\boxed{\psi(\nu - \nu_0) = \frac{\gamma/4\pi^2}{(\nu - \nu_0)^2 + (\gamma/4\pi)^2}} \qquad \int_0^\infty \psi(\nu - \nu_0) d\nu = 1$$

The full width at half maximum (FWHM) is given by

$$\text{FWHM} = \gamma/2\pi$$

- **In a two-level atom, the single transition has**

$$\gamma = \gamma_u = A_{ul}$$

because the lower level of a two-level atom has infinite lifetime and does not contribute existential uncertainty.

- [Derivation]

See "Introduction to Quantum Mechanics (Brandsen and Joachain, 1989)" and "Radiative Processes in Astrophysics (Rybicki & Lightman, 2004)

The profile describes the convolution of two Fourier transforms.

- The first is the delta function $\delta(\nu - \nu_0)$ that represents the Fourier transform of an infinitely long, monochromatic wave $\exp(-i2\pi\nu_0 t)$ with frequency ν_0.
- The second is the Fourier transform of the $\exp(-A_{ul}t) = \exp(-\gamma t)$ decay distribution.
- The energy amplitude distribution is given by the Fourier transform of the convolution of the two separate component transforms.

Semiclassical (Weisskopf-Woolley) Picture of Quantum Levels

- In the semiclassical picture, each level is viewed as a continuous distribution of sublevels with energies close to the energy of the level (E_n).

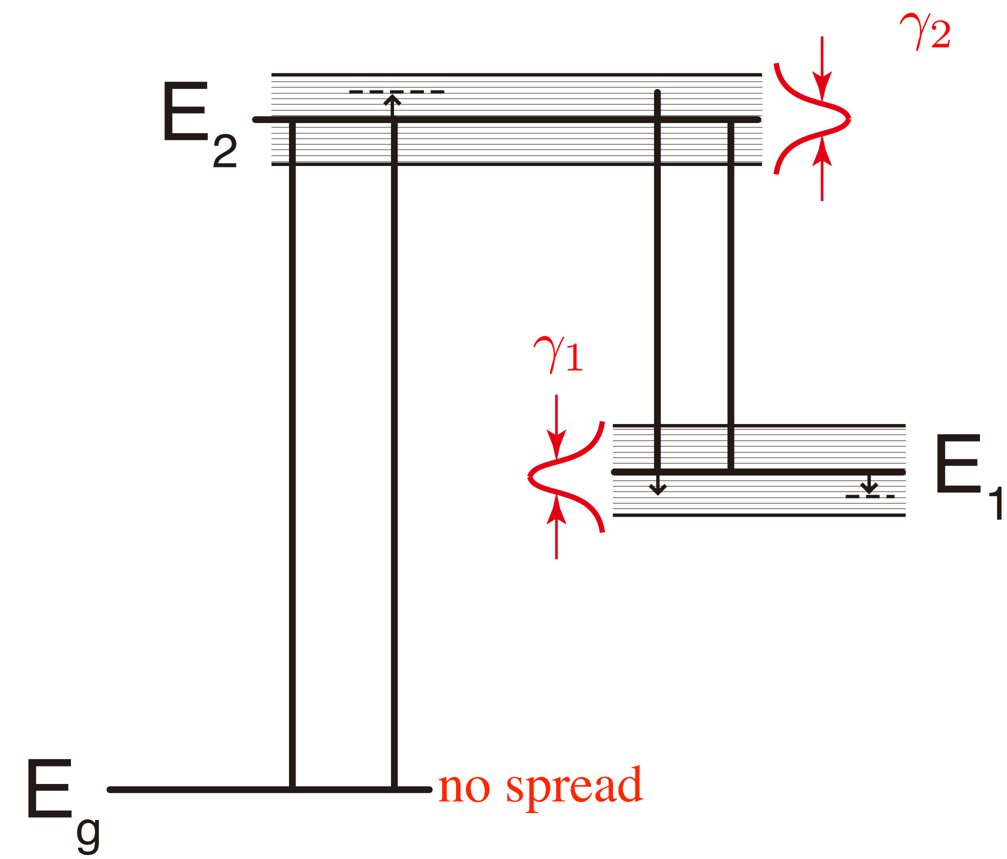
  The distribution of sublevels are explained by the Heisenberg Uncertainty Principle. The level has a lifetime $\Delta t = 1/A$ (A = Einstein A coefficient) and a spread in energy about $\Delta E \approx \hbar/\Delta t = \hbar A$.

$$\Delta E \Delta t \approx \hbar$$

The ground level has no spread in energy

because $\Delta t = \infty$.

The atom is in a definite sublevel of some level.



A transition in a spectral line is considered to be an **instantaneous transition** between a definite sublevel of an initial level to a definite sublevel of a final level.

- ***The energy spread of sublevels is described by a Lorentzian profile with the damping parameter of $\gamma = A$.***
- ***This picture implies that the emission line profile is the same as the absorption line profile.***

- Multiple levels and transitions.

In real atoms and ions the lower level of a given line may also have finite lifetime (when it isn't the ground state) and there may also be multiple downward transitions from each level. Multiple transition probabilities add up as

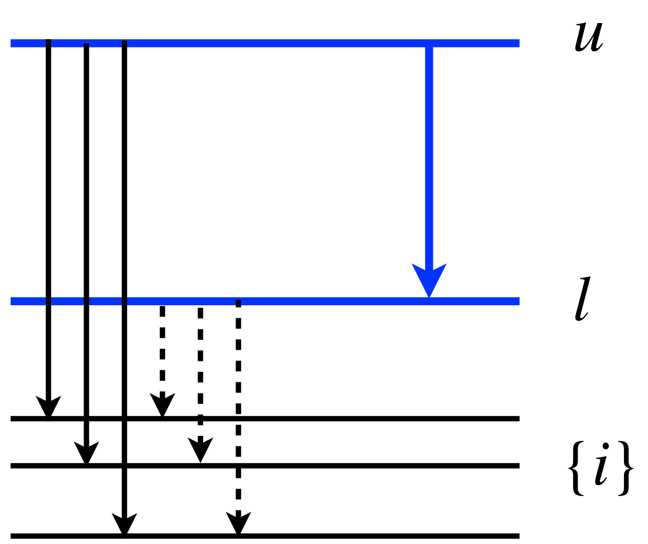
$$\gamma_u = \sum_{l<u} A_{ul}$$

This is because the corresponding decay functions multiply as $\exp(-A_{u1}t)\exp(-A_{u2}t) = \exp[-(\gamma_1 + \gamma_2)t]$ so that the γ broadening parameters add linearly.

In other words, the convolution of two Lorentz profiles with halfwidth parameters γ_1 and γ_2 gives a new Lorentz profile with halfwidth parameter $\gamma = \gamma_1 + \gamma_2$.

The total natural damping width is therefore given by:

$$\boxed{\gamma = \gamma_l + \gamma_u = \sum_{i<l} A_{li} + \sum_{i<u} A_{ui}}$$



Collision Broadening

- **Elastic collisions**

  Collisional broadening or pressure broadening results from other particles in the neighborhood. They may be electrons, ions, atoms or molecules.

  Their charge affects the radiating or extincting atom or ion of interest through the Coulomb interaction and therefore affects the frequency of a bound-bound transition between perturbed levels.

  These collisional encounters are often termed ***elastic*** although the energies are slightly changed momentarily.

  The term ***inelastic*** is reserved for collisions involving bound-bound transitions between different energy levels.

- **Impact approximation**

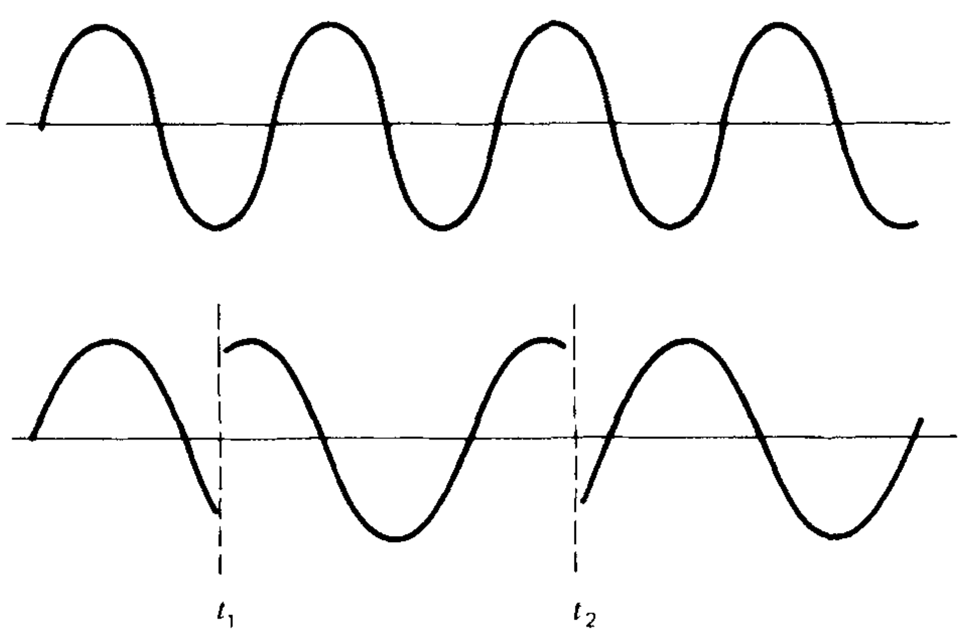
  The perturber comes by at large speed and causes a momentary disruption of the wave train emitted by a deexciting atom.

  Such disruptions are described as phase jumps in the wave train (Lindholm theory) and cause broadening with a Lorentz profile just as the natural-broadening decay functions do.

$$\boxed{\psi(\nu-\nu_0)=\frac{\Gamma/4\pi^2}{(\nu-\nu_0)^2+(\Gamma/4\pi)^2}\quad \text{where}\quad \Gamma=\gamma+2\nu_{\mathrm{col}}}$$

Here, ν_{col} = collisions per unit time

The impact approximation holds at low densities where the duration of collision is much shorter than the time between collisions.



Redistribution Function

- Redistribution Function:

  The redistribution function describes any correlation between an absorbed and emitted photon during the scattering in a spectral line.

  Frequency Redistribution Function $r(\nu', \nu)d\nu' d\nu$ = the joint probability density that a photon in the frequency range $\nu', \nu' + d\nu'$ is absorbed, and subsequently a photon in the frequency range ν, $\nu + d\nu$ is emitted, in the transition $i \to e \to f$.

- **Completely Correlated Scattering / Coherent Scattering (Coherent Redistribution)**

  A limiting case where the frequency of the emitted photon is exactly equal to the frequency of the absorbed photon in the atom's rest frame, or, in other words, in which an atom does not experience elastic collisions while it is in the excited level, and there is no reshuffling of photons to other sublevels within level e.

- **Completely Uncorrelated Scattering / Complete Frequency Redistribution (CFR)**

  A limiting case where the frequency of the emitted photon is completely independent of the frequency of the absorbed photon.

  It corresponds to complete reshuffling of the excited electrons overlapping all the sublevels of level e, so that re-emission of a photon from level e is proportional to the number of sublevels at the level e.

$$r^{\mathrm{uncorr}}(\nu', \nu) = \psi(\nu')\psi(\nu)$$

- **Partial Frequency Redistribution (PFR)**

The general redistribution function for a mixture of some reshuffling and some no-reshuffling in the excited state.

In practice, the term PFR may refer to any kind of departure from complete redistribution, including the case of full coherence.

The general redistribution function is written as

$$r^{iji}(\nu', \nu) = p_j^{\mathrm{corr}} r_{iji}^{\mathrm{corr}} + (1 - p_j^{\mathrm{corr}}) r_{iji}^{\mathrm{uncorr}}(\nu', \nu)$$

Here,

$$p_j^{\mathrm{corr}} = \frac{P_j}{P_j + Q_j}$$

P_j = the total rate of transition out of level j with elastic collisions omitted.

Q_j = the rate of elastic collisions, i.e., transitions to another sublevel of the same level.

p_j^{corr} = the probability that a photon is re-emitted from the same level (before an elastic collision). This quantity is also called the ***coherence fraction***.

Doppler Broadening

Doppler shift. The motion of a radiating particle along the line of sight produces a Doppler shift given by (for $\xi \ll c$):

$$\frac{\Delta\nu}{\nu} = -\frac{\Delta\lambda}{\lambda} = \frac{\xi}{c} \tag{3.52}$$

with ξ the velocity component along the line of sight. Its sign is usually taken positive towards the observer, so that positive Doppler shift means blueshift and upward motion in the stellar atmosphere. A photon that is emitted at frequency ν' in the frame of the emitting atom is detected by the observer as blueshifted to the frequency ν with

$$\nu = \nu'(1+\xi/c) \approx \nu' + \nu_0\xi/c \tag{3.53}$$

where ν' is replaced by the line-center frequency ν_0 in the second term since $\xi/c \ll 1$. The same result is obtained for an absorbing atom that moves with velocity ξ towards the observer and extincts radiation that it sees redshifted in its own frame to

$$\nu' = \nu(1-\xi/c) \approx \nu - \nu_0\xi/c. \tag{3.54}$$

Thermal motions. For purely thermal motions the distribution of velocities in the line of sight is given by the component form of the Maxwell distribution:

$$\frac{n(\xi)}{N}\,\mathrm{d}\xi = \frac{1}{\xi_0\sqrt{\pi}}\,\mathrm{e}^{-\xi^2/\xi_0^2}\,\mathrm{d}\xi \tag{3.55}$$

which is an area-normalized Gaussian distribution with variance

$$\xi_0 = \sqrt{\frac{2kT}{m}} = 12.85\sqrt{\frac{T}{10^4 A}} \tag{3.56}$$

with k the Boltzmann constant, m the mass of the atom, A the atomic weight and ξ_0 in km s^{-1}. The root-mean-square velocity component in the line of sight is:

$$<\xi^2>^{1/2} = \sqrt{\frac{kT}{m}} = \frac{\xi_0}{\sqrt{2}}. \tag{3.57}$$

Thermal broadening. What is the line extinction profile when thermal motions are taken into account? Remember that the monochromatic extinction coefficient per particle σ^l_ν in (2.65) on page 23 is an ensemble probability, representing the average value for all particles that may cause extinction through that bound-bound transition at that wavelength. Write it as $\sigma^l_\nu \equiv \sigma^l(\nu - \nu_0)$. The total over the line is given by

$$\sigma^l_{\nu_0} \equiv \int_0^\infty \sigma^l(\nu - \nu_0)\,\mathrm{d}\nu = \int_0^\infty \frac{h\nu}{4\pi} B_{lu} \varphi(\nu - \nu_0)\,\mathrm{d}\nu = \frac{\pi e^2}{m_\mathrm{e} c} f \int_0^\infty \varphi(\nu - \nu_0)\,\mathrm{d}\nu = \frac{\pi e^2}{m_\mathrm{e} c} f \quad (3.58)$$

with $f \equiv f_{lu}$ the classical oscillator strength in (2.66). The ensemble-averaged extinction coefficient per particle in the particle frame is per line-of-sight velocity ξ using (3.54):

$$\sigma^l_{\nu'} = \sigma^l(\nu' - \nu_0) = \sigma^l(\nu - \xi\nu_0/c - \nu_0). \quad (3.59)$$

This must be averaged over the velocity distribution $n(\xi)$ to obtain the ensemble-averaged extinction coefficient in the frame of the observer:

$$\sigma^l(\nu - \nu_0) = \frac{\int \sigma^l(\nu - \xi\nu_0/c - \nu_0)\, n(\xi)\,\mathrm{d}\xi}{\int n(\xi)\,\mathrm{d}\xi} = \int_{-\infty}^{+\infty} \sigma^l(\nu - \xi\nu_0/c - \nu_0)\,\frac{n(\xi)}{N}\,\mathrm{d}\xi \quad (3.60)$$

which represents a convolution (see footnote on page 54). In the case that thermal Doppler shifts constitute the *only* broadening[5] the local extinction profile shape per particle $\varphi(\nu' - \nu_0)$ is replaced by a delta function $\delta(\nu' - \nu_0)$ because in that case both levels are sharp and each extinction concerns a single (shifted) frequency only. The convolution is then, using (2.66) on page 23,

$$\sigma^l(\nu - \nu_0) = \frac{\pi e^2}{m_\mathrm{e} c} f \int_{-\infty}^{+\infty} \delta(\nu - \xi\nu_0/c - \nu_0)\,\frac{n(\xi)}{N}\,\mathrm{d}\xi \quad (3.61)$$

and produces the monochromatic extinction coefficient in the frame of the observer, measured per particle but averaged over all line-of-sight particle velocities, by setting $\xi = (\nu - \nu_0)\,c/\nu_0$ in the Maxwell distribution $n(\xi)/N$:

$$\sigma^l_\nu = \frac{\pi e^2}{m_\mathrm{e} c} f \, \frac{n\left[(\nu - \nu_0)\,c/\nu_0\right]}{N} = \frac{\sqrt{\pi} e^2}{m_\mathrm{e} c} \frac{f}{\Delta\nu_\mathrm{D}}\, \mathrm{e}^{-(\Delta\nu/\Delta\nu_\mathrm{D})^2} \quad (3.62)$$

with the *Doppler width* $\Delta\nu_\mathrm{D}$ defined by

$$\Delta\nu_\mathrm{D} \equiv \frac{\xi_0}{c}\nu_0 = \frac{\nu_0}{c}\sqrt{\frac{2kT}{m}} \quad (3.63)$$

and the Gaussian extinction profile given by

$$\varphi(\nu - \nu_0) = \frac{1}{\sqrt{\pi}\,\Delta\nu_\mathrm{D}}\, \mathrm{e}^{-(\Delta\nu/\Delta\nu_\mathrm{D})^2}. \quad (3.64)$$

Voigt profile. When the collisional damping has the Lorentz profile shape (as in the impact approximation), the total damping profile is the convolution of the constituents and is given by a Lorentz profile with $\gamma = \gamma^{\rm rad} + \gamma^{\rm col}$. Assuming that collisional broadening and thermal Doppler shifting are independent processes, this shape function must be convolved with (3.62) for thermal broadening to obtain the total extinction coefficient. The damping profile (3.44) is area-normalized and therefore takes the place of the δ function in (3.61), again with the frame-of-the-atom frequency shift $\nu' - \nu_0 = \nu - \xi\nu_0/c - \nu_0$ as argument. Thus:

$$\begin{aligned}
\sigma^l_\nu &= \left[\frac{\sqrt{\pi}e^2}{m_{\rm e}c}\frac{f}{\Delta\nu_{\rm D}}\,{\rm e}^{-(\Delta\nu/\Delta\nu_{\rm D})^2}\right] * \left[\frac{\gamma/4\pi^2}{(\nu'-\nu_0)^2 + (\gamma/4\pi)^2}\right] \\
&= \frac{\sqrt{\pi}e^2}{m_{\rm e}c}\frac{f}{\Delta\nu_{\rm D}}\int_{-\infty}^{+\infty}\frac{(\gamma/4\pi^2)\,{\rm e}^{-(\Delta\nu/\Delta\nu_{\rm D})^2}}{(\nu'-\nu_0)^2 + (\gamma/4\pi)^2}\,{\rm d}\nu \\
&= \frac{\sqrt{\pi}e^2}{m_{\rm e}c}\frac{f}{\Delta\nu_{\rm D}}H(a,v) \qquad (3.67)
\end{aligned}$$

with

$$\begin{aligned}
H(a,v) &\equiv \frac{a}{\pi}\int_{-\infty}^{+\infty}\frac{{\rm e}^{-y^2}}{(v-y)^2 + a^2}\,{\rm d}y \qquad (3.68)\\
y &\equiv \frac{\xi}{\xi_0} = \frac{\xi}{c}\frac{\nu_0}{\Delta\nu_{\rm D}} = \frac{\xi}{c}\frac{\lambda_0}{\Delta\lambda_{\rm D}} \qquad (3.69)\\
v &\equiv \frac{\nu-\nu_0}{\Delta\nu_{\rm D}} = \frac{\lambda-\lambda_0}{\Delta\lambda_{\rm D}} \qquad (3.70)\\
a &\equiv \frac{\gamma}{4\pi\Delta\nu_{\rm D}} = \frac{\lambda^2}{4\pi c}\frac{\gamma}{\Delta\lambda_{\rm D}}. \qquad (3.71)
\end{aligned}$$

The function $H(a,v)$ is called the *Voigt function.* Examples are shown in Fig. 3.1; a simple

The Voigt function is not normalized but has area $\sqrt{\pi}$ in v units; the maximum value in the center has $H(a, v=0) \approx 1-a$ for $a < 1$. The corresponding area-normalized extinction profile is:

$$\varphi(\nu-\nu_0) = \frac{H(a,v)}{\sqrt{\pi}\,\Delta\nu_{\rm D}}. \qquad (3.72)$$

The rough approximation for $a \ll 1$

$$H(a,v) \approx {\rm e}^{-v^2} + \frac{a}{\sqrt{\pi}\,v^2} \qquad (3.73)$$

shows that the shape of the Voigt function approximates a Gaussian near line center but possesses $\Delta\nu^{-2}$ damping decay in the far wings

Plasma Effects

Introduction

- Light travels slower in a medium than in a vacuum because photons constantly interact with the atoms and electrons of the material, causing a time-lag in absorption and re-emission that lowers the overall propagation speed.
  - The oscillating electric field of the light wave forces electrons in the material to oscillate, creating their own electromagnetic field. The interference between the original light wave and the fields created by the oscillating electrons slows down the overall wavefront.
  - The amount of slowdown depends on the medium's density and optical properties.

- **Dispersion**: The electrons and magnetic fields of the ionized interstellar medium (ISM) alter the character of radio waves propagating through it. The effect of the electron is to slow the propagation speed of the low-frequency components of a signal relative to that of the higher frequencies. **This phenomenon, the variation of transmission speed with frequency through a plasma, is known as dispersion.**
  - A pulse of radiation emitted by a radio pulsar will contain a wide range of radio frequencies (e.g., 100-400 MHz). During the pulse's passage through the ISM, the higher frequencies propagate faster than the lower ones. A receiver on earth tuned to 100 MHz will detect the pulse substantially delayed relative to the arrival time at 400 MHz.
  - Radio pulsars emit sequential pulses of radiation, and so the dispersion is repeated again and again. One could not observe dispersion with a steady source; all frequencies would arrive at all times.

- Roughly speaking a **plasma** is a globally neutral partially or completely ionized gas.

[Dispersion in Cold, Isotropic Plasma]

- Maxwell's equations for a vacuum can be used for a plasma if the charge and current densities ρ and $\mathbf{j}$ due to the plasma are explicitly included.
- <u>**Plasma Dispersion Relation**</u>

If we assume a space and time variation of all quantities of the form $\exp\left[i\left(\mathbf{k}\cdot\mathbf{r}-\omega t\right)\right]$, Maxwell's equations become

Gauss's law	$\nabla\cdot\mathbf{E}=4\pi\rho$
Gauss's law for magnetic field	$\nabla\cdot\mathbf{B}=0$
Ampere's law	$\nabla\times\mathbf{E}=-\frac{1}{c}\frac{\partial\mathbf{B}}{\partial t}$
Faraday's law	$\nabla\times\mathbf{B}=\frac{4\pi}{c}\mathbf{j}+\frac{1}{c}\frac{\partial\mathbf{E}}{\partial t}$

$\longrightarrow$

$$
\boxed{
\begin{aligned}
i\mathbf{k}\cdot\mathbf{E}&=4\pi\rho\\
i\mathbf{k}\cdot\mathbf{B}&=0\\
i\mathbf{k}\times\mathbf{E}&=i\frac{\omega}{c}\mathbf{B}\\
i\mathbf{k}\times\mathbf{B}&=\frac{4\pi}{c}\mathbf{j}-i\frac{\omega}{c}\mathbf{E}
\end{aligned}
}
$$

Let us assume that the our plasma consists of electrons with density n. The ions are very much less mobile than electrons and thus are neglected. We also assume that there is no external magnetic field; thus the plasma is isotropic. Then, the equation of motion of electrons when there is no external magnetic field is

$$m_e\dot{\mathbf{v}}=-e\left(\mathbf{E}+\cancel{\frac{\mathbf{v}}{c}\times}\mathbf{B}\right)=-e\mathbf{E}\quad\longrightarrow\quad
\begin{aligned}
-im_e\omega\mathbf{v}&=-e\mathbf{E}\\
\mathbf{v}&=\frac{e\mathbf{E}}{i\omega m_e}
\end{aligned}$$

<u>Current density:</u> $\mathbf{j}=-ne\mathbf{v}=\sigma\mathbf{E}$ where the $\boxed{\textbf{conductivity is } \sigma\equiv\frac{ine^2}{\omega m_e}.}$

The charge conservation equation (continuity equation) gives

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0 \qquad -i\omega\rho + i\mathbf{k}\cdot\mathbf{j} = 0 \;\rightarrow\; \rho = \frac{1}{\omega}\mathbf{k}\cdot\mathbf{j} = \frac{\sigma}{\omega}\mathbf{k}\cdot\mathbf{E}$$

Using the expressions for **j** and ρ, Maxwell's equations become

$$\begin{aligned} \mathbf{j} &= \sigma\mathbf{E} \\ \rho &= \frac{\sigma}{\omega}\mathbf{k}\cdot\mathbf{E} \end{aligned} \longrightarrow \begin{aligned} i\mathbf{k}\cdot\mathbf{E} &= 4\pi\rho \\ i\mathbf{k}\cdot\mathbf{B} &= 0 \\ i\mathbf{k}\times\mathbf{E} &= i\frac{\omega}{c}\mathbf{B} \\ i\mathbf{k}\times\mathbf{B} &= \frac{4\pi}{c}\mathbf{j} - i\frac{\omega}{c}\mathbf{E} \end{aligned} \longrightarrow \begin{aligned} i\mathbf{k}\cdot\epsilon\mathbf{E} &= 0 \\ i\mathbf{k}\cdot\mathbf{B} &= 0 \\ i\mathbf{k}\times\mathbf{E} &= i\frac{\omega}{c}\mathbf{B} \\ i\mathbf{k}\times\mathbf{B} &= -i\frac{\omega}{c}\epsilon\mathbf{E} \end{aligned}$$

These equations are now "source-free" and can be solved.

We find that $\mathbf{k},\ \mathbf{E},\ \mathbf{B}$ form a mutually orthogonal right-hand vector triad.

The **dispersion relation** connecting k and ω is given by

$$\begin{aligned} i\mathbf{k}\times\mathbf{B} &= -i\frac{\omega}{c}\epsilon\mathbf{E} \\ i\mathbf{k}\times\left(\frac{c}{\omega}\mathbf{k}\times\mathbf{E}\right) &= -i\frac{\omega}{c}\epsilon\mathbf{E} \\ \mathbf{k}\left(\cancel{\mathbf{k}\cdot\mathbf{E}}\right) - \mathbf{E}\left(\mathbf{k}\cdot\mathbf{k}\right) &= -\frac{\omega^2}{c^2}\epsilon\mathbf{E} \end{aligned} \longrightarrow \boxed{\begin{aligned} c^2k^2 &= \epsilon\omega^2 \\ &= \omega^2 - \omega_p^2 \\ \omega^2 &= \omega_p^2 + c^2k^2 \end{aligned}}$$

where the **dielectric constant**:

$$\epsilon \equiv 1 - \frac{4\pi\sigma}{i\omega} = 1 - \frac{4\pi n e^2}{m_e\omega^2} = 1 - \left(\frac{\omega_p}{\omega}\right)^2$$

plasma frequency:

$$\omega_p \equiv \sqrt{\frac{4\pi n e^2}{m_e}} = 5.63\times 10^4\left(n/\mathrm{cm}^{-3}\right)^{1/2}\ \mathrm{s}^{-1}$$

- We now see immediately that **for $\omega < \omega_p$, the wavenumber is imaginary** ($k^2 < 0$).

$$k = \frac{i}{c}\sqrt{\omega_p^2 - \omega^2} \qquad\qquad e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)} \propto e^{-|k|r}$$

In this case, the wave amplitude decreases as $e^{-|k|r}$ on a scale of the order of $1/|k| \approx c/\omega_p$.

Thus the plasma frequency ω_p defines **a plasma cutoff frequency below which there is no electromagnetic propagation**.

For instance, Earth ionosphere prevents extraterrestrial radiation at frequencies less than ~ 1 MHz from being observed at the earth's surface ($n \sim 10^4$ cm^{-3}). The existence of the plasma cutoff yields an important method of probing the ionosphere.

Radio astronomy from the earth's surface can not be carried out at frequencies $\nu \lesssim 9$ MHz. Such radio waves from outer space are reflected back to space by the ionosphere.

Method of probing the ionosphere:

Let a pulse of radiation in a narrow range about ω be directed straight upward from the earth's surface.

When there is a layer at which the density n is large enough to make $\omega < \omega_p$, the pulse will be totally reflected from the layer.

The time delay of the pulse provides information on the height of the layer.

By making such measurements at many different frequencies, the electron density can be determined as a function of height.

- **Radio Broadcast**

  AM radio (~ 500-1700 kHz) can propagate long distances via such reflections from the bottom of the ionosphere. This is particularly effective at night when the relatively low “D” layer of the ionosphere becomes neutral (and disappears); then reflection can take place at higher altitudes that have large values of n_e.

  In contrast, FM radio at ~ 100 MHz travels right through the ionosphere into outer space; it sounds better, but one has to be close enough to the station that the signal can be received more or less directly without ionosphere reflections.

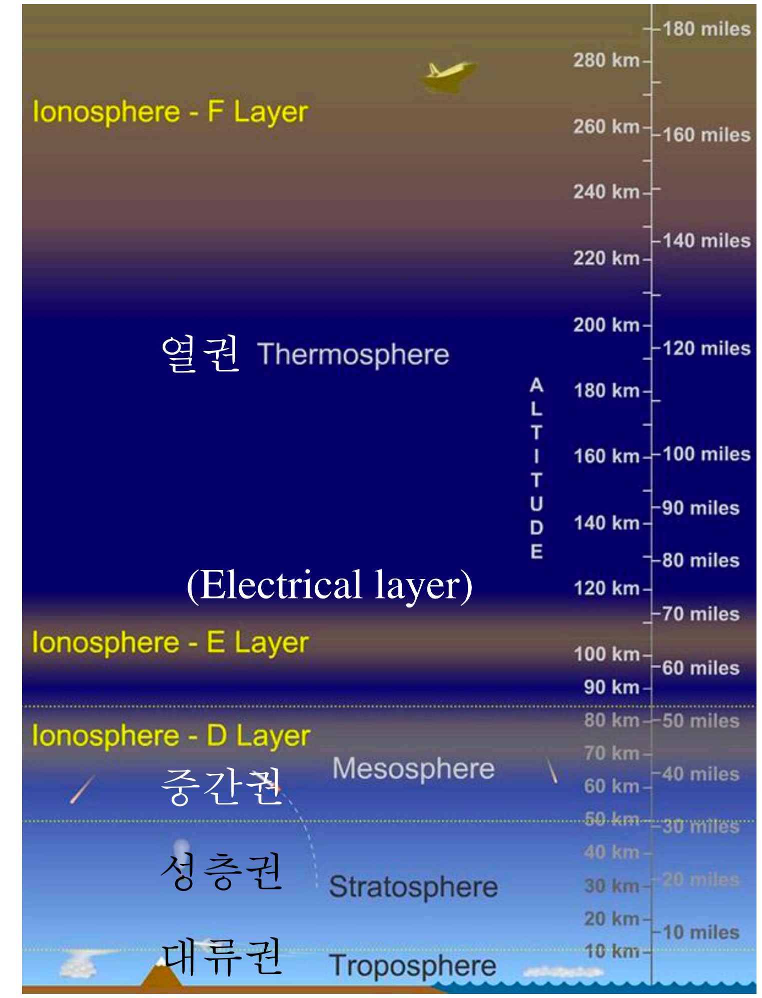
  The ionosphere is between 80 and ~ 600 km where EUV and X-ray solar radiation ionize the atoms and molecules thus creating a layer of electrons.

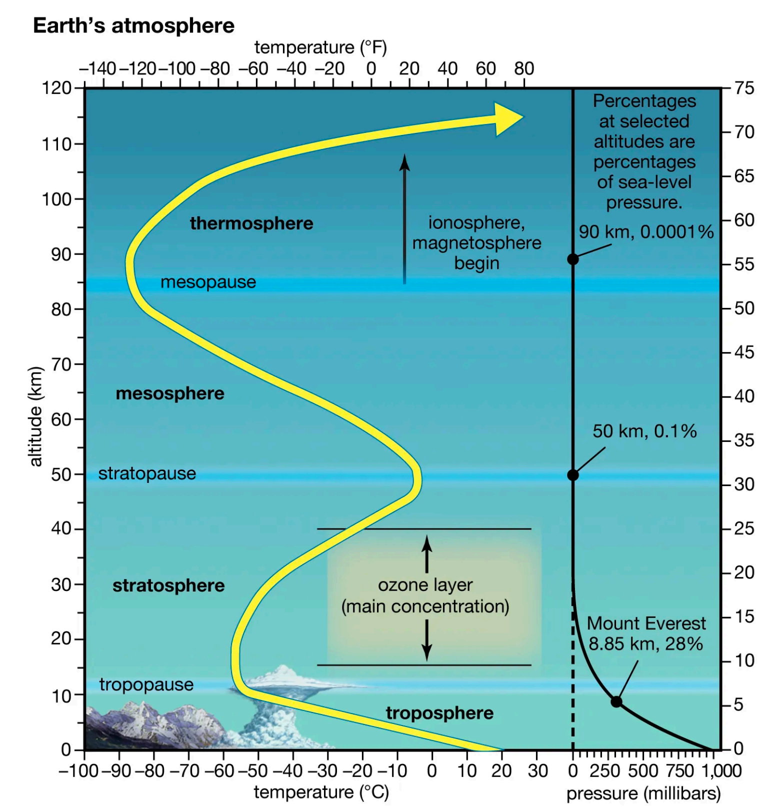
- **Interstellar cutoff**

  In interstellar space, the number densities of ions are typically much less ($n_e \sim 1\ \mathrm{cm}^{-3}$); thus

$$\nu_p \approx 300 - 1000\,\mathrm{Hz}$$

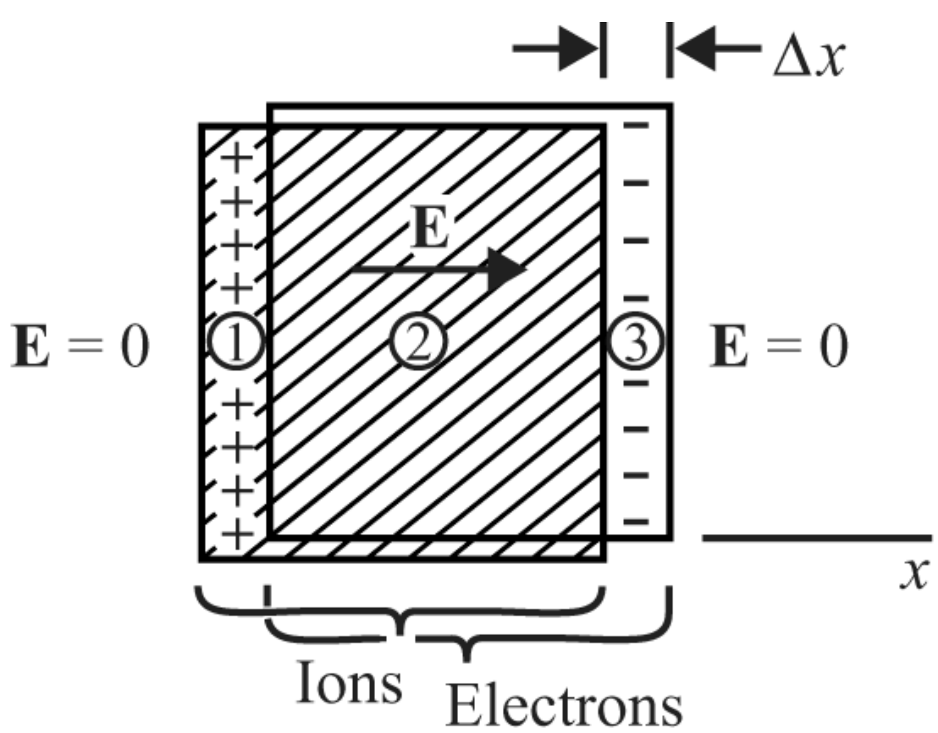
  This is well below the frequencies that penetrate the ionosphere ($\nu_p \gtrsim 9\,\mathrm{MHz}$). Consequently, **radio waves that can penetrate the atmosphere can easily travel through much of interstellar space. This makes galactic and extragalactic radio astronomy possible.**





[Plasma Frequency]

- If the electrons in a uniform, homogeneous plasma are displaced from their equilibrium position, an electric field arises because of charge separation. This electric field produces a restoring force on the displaced electrons. Since the magnitude of the charge imbalance is directly proportional to the displacement, the restoring force is given by Hooke's law, $F = -k\Delta x$ and the system behaves as a harmonic oscillator. The resulting oscillations are called **electron plasma oscillations**. The oscillation frequency is called the **electron plasma frequency**.



$$\nabla \cdot \mathbf{E} = 4\pi\rho \quad \rightarrow \quad E = 4\pi n e \Delta x$$

n = electron density

$$m_e \frac{d^2 \Delta x}{dt^2} = -eE = -4\pi n e^2 \Delta x$$

$$\frac{d^2 \Delta x}{dt^2} + \frac{4\pi n e^2}{m_e} \Delta x = 0$$

$$\frac{d^2 \Delta x}{dt^2} + \omega_p^2 \Delta x = 0$$

$$\text{where} \quad \omega_p \equiv \frac{4\pi n e^2}{m_e}$$

[Phase Velocity and Group Velocity]

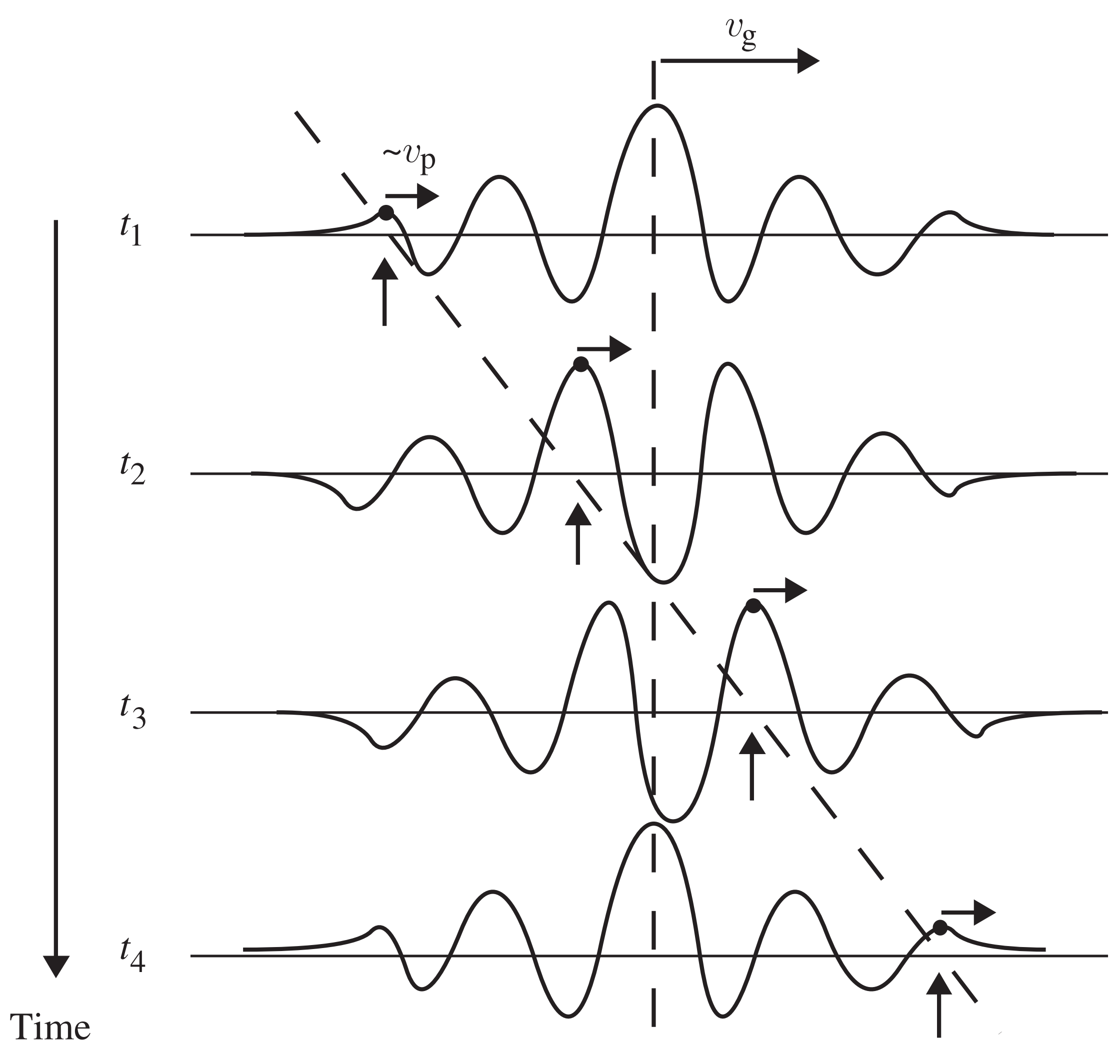
- Dispersion relation, phase velocity, and group velocity:

dispersion relation = a function which gives ω as a function of k.

phase velocity = the rate at which the phase of the wave propagates in space.

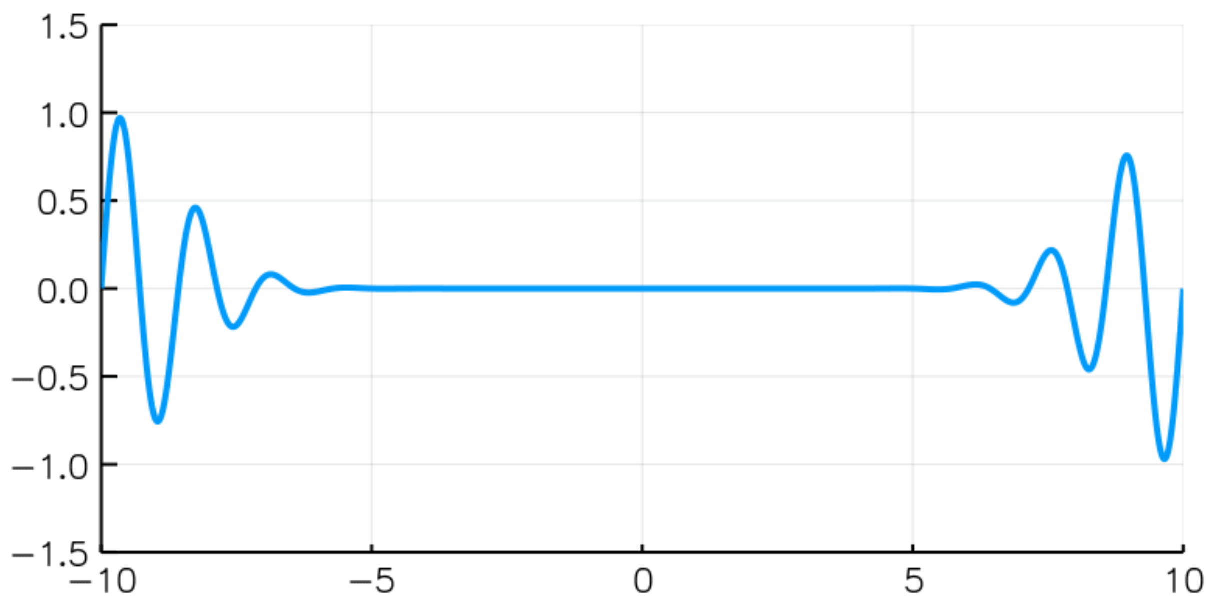
group velocity = the velocity with which the overall shape of the waves' amplitudes (modulation or envelope of the wave) propagates through space.

$$\boxed{\text{In vacuum,} \quad \omega = ck, \quad v_{\rm ph} \equiv \frac{\omega}{k} = c, \quad v_g \equiv \frac{\partial \omega}{\partial k} = c}$$



A wavelet or wave packet shown at four different times. The packet as a whole moves to the right at group velocity v_g. The interference between the component waves that make up the packet, with their different phase velocities, causes a single crest (arrows and dots) to gradually overtake the centroid of the packet. Relative to the medium, the overtaking crest moves at approximately the median phase velocity.

Figure 11.3. [Bradt] Astrophysics Processes



https://en.wikipedia.org/wiki/Group_velocity

- Assume that a wave packet E is almost monochromatic, so that its Fourier component is nonzero only in the vicinity of a central wavenumber k_0. Then, linearization gives:

$$\omega(k) \approx \omega_0 + (k - k_0) \left. \frac{\partial \omega}{\partial k} \right|_{k=k_0} = \omega_0 + (k - k_0) v_g$$

$$\longrightarrow$$

$$E(x,t) = \int dk \int d\omega \bar{E}(k,\omega) e^{i(kx-\omega t)} \approx e^{it(v_g k_0 - \omega_0)} \int dk \bar{E}(k,\omega_0) e^{ik(x - v_g t)}$$

where $\boxed{v_g \equiv \left. \frac{\partial \omega}{\partial k} \right|_{k=k_0}}$

$$\boxed{|E(x,t)| = |E(x - v_g t, 0)|}$$

- Therefore, the envelope of the wavepacket travels at velocity $v_g = \left(\partial\omega/\partial k\right)_{k=k_0}$.

- **Group and Phase Velocity in terms of the index of refraction**

Phase velocity

$$\boxed{v_{\text{ph}} \equiv \frac{\omega}{k} = \frac{\omega}{\sqrt{\omega^2 - \omega_p^2}/c} = \frac{c}{n_r}}$$

where ***the index of refraction*** is defined to be $\boxed{n_r \equiv \sqrt{\epsilon} = \sqrt{1 - \omega_p^2/\omega^2}}$

When $\omega > \omega_p$, waves do propagate without damping with phase velocity and *the phase velocity always exceeds the speed of light* $\left(v_{\text{ph}} > c\right)$.

Group velocity

Group velocity is the speed with which a pulse of radiation containing a band of frequencies will propagate through space.

$$\omega^2 = \omega_p^2 + c^2k^2 \quad \longrightarrow \quad \boxed{v_{\text{g}} \equiv \frac{\partial\omega}{\partial k} = \frac{c^2k}{\omega} = c\sqrt{1 - \frac{\omega_p^2}{\omega^2}} \quad \rightarrow \quad v_{\text{g}} < c}$$

$$v_{\text{ph}} = \frac{c}{n_r}$$
$$v_{\text{g}} = cn_r$$

The group velocity is always less than c. The wave energy travels at the group velocity, as does any modulation of the wave (information coding). **The information content moves at the group velocity.**

The lower frequencies travel more slowly than higher frequencies. For $\omega < \omega_p$, the index of refraction becomes imaginary, leading to attenuation. If the frequency ω is lowered toward ω_p and increasing, the group velocity decreases toward zero. If the frequency ω is well above ω_p and increasing, the group velocity approaches c. As the density is lowered ($n \rightarrow 0$; become more closely a vacuum), ω_p decreases and the group velocity more closely approaches the vacuum velocity c.

[Dispersion Measure]

- **Pulsar**

An important application of the formula for group velocity is to pulsars. Each individual pulse from the pulsar has a spectrum covering a wide band of frequency. Thus, the pulse will be dispersed by its interaction with the interstellar plasma, since each small range of frequencies travels at a slightly different group velocity and will reach earth at a slightly different time.

Let d = pulsar distance to the pulsar

The time required for a pulse to reach earth at frequency ω is $t_p = \int_0^d \frac{ds}{v_g}$.

The plasma frequency in ISM is usually quite low (~ 10^3 Hz), so we can assume that $\omega \gg \omega_p$. Then, the travel time is given by

$$\frac{1}{v_g} = \frac{1}{c}\left(1 - \frac{\omega_p^2}{\omega^2}\right)^{-1/2} \approx \frac{1}{c}\left(1 + \frac{1}{2}\frac{\omega_p^2}{\omega^2}\right) \longrightarrow \boxed{t_p \approx \frac{d}{c} + \frac{2\pi e^2}{c m_e \omega^2}\int_0^d n\,ds}$$

= transit time for a vacuum + plasma correction

- **Dispersion measure**:

What is usually measured is **the rate of change of arrival time with respect to frequency**.

$$\boxed{\frac{dt_p}{d\omega} = -\frac{4\pi e^2}{c m_e \omega^3}\mathcal{DM} \quad \text{where} \quad \mathcal{DM} \equiv \int_0^d n\,ds}$$

is the dispersion measure of the ray.

This is the electron column density.

The arrival interval of pulsar signal between two frequencies ν_1 and ν_2:

$$\boxed{\Delta t_p = \frac{e^2}{2\pi c m_e}\left(\frac{1}{\nu_1^2} - \frac{1}{\nu_2^2}\right)\mathcal{DM} = 4.15 \times 10^{15}\left(\frac{1}{\nu_1^2} - \frac{1}{\nu_2^2}\right)\frac{\mathcal{DM}}{\mathrm{cm^{-3}\ pc}}\ \mathrm{sec}}$$

- **Crab nebula**

Observations of the pulses from the Crab pulsar show that the delay between frequencies 100 MHz and 400 MHz is 23.5 s, which yields a value for the column density expressed as

$$\int_0^D n_e ds = 1.86 \times 10^{20}\,\mathrm{cm}^{-2}. \quad \text{electron column density to Crab pulsar}$$

Note that astronomers use the mixed units pc cm^{-3} to denote the dispersion measure (DM), but cm^{-2} for the column density.

Because the distance to the Crab pulsar is about 2 kpc (= 6.2×10^{21} cm), the average electron density is

$$\langle n_e \rangle_{\mathrm{abs}} = \frac{1}{D}\int_0^D n_e ds = 0.03 \text{ electrons cm}^{-3}$$

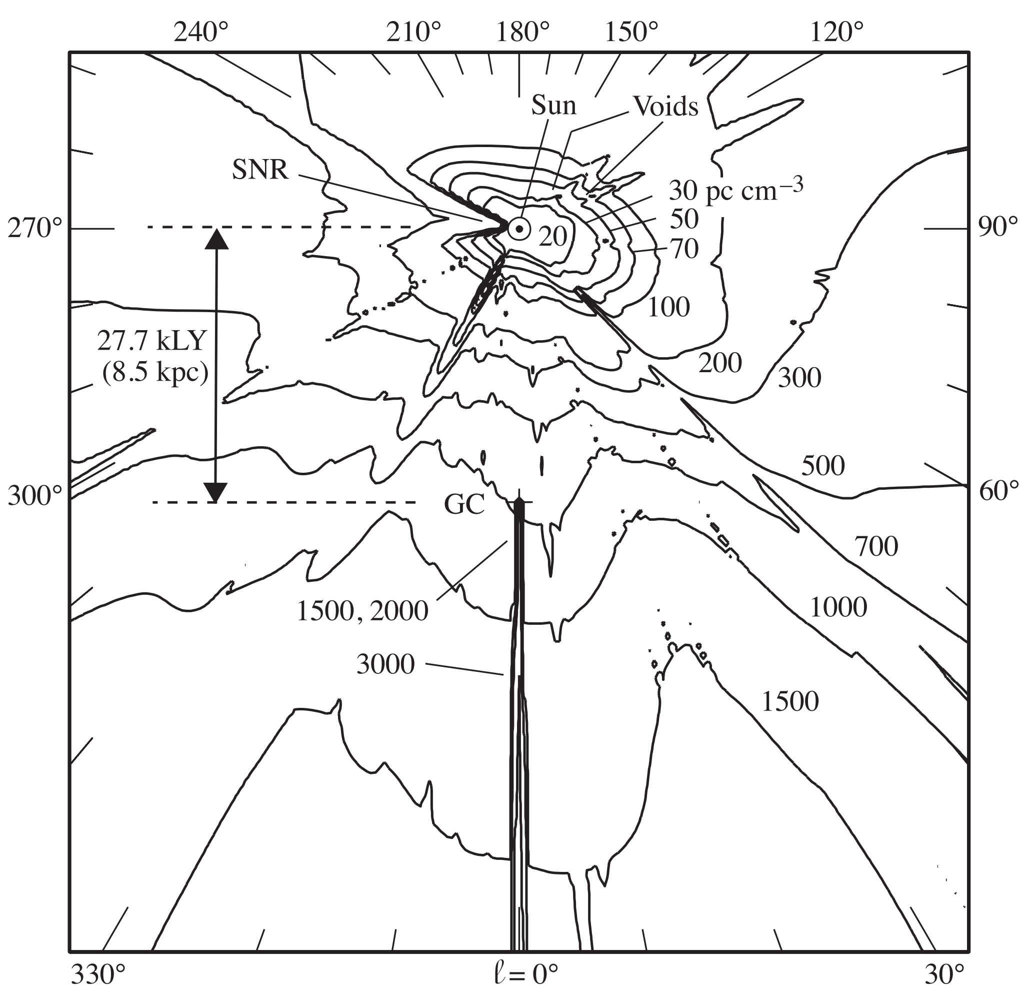
This is relatively low density in the context of the ISM. It lies between the densities of the Warm Ionized Medium (WIM) and Hot Ionized Medium (HIM).

The 23-s time lag from 400 MHz to 100 MHz is very long compared with the 33-ms interval between pulses from the Crab pulsar. At earth, each pulse is “heard” as a frequency that continuously decreases.

- **Galactic model of electron density**

The number of known radio pulsars is now over 2000, and the DM has been measured for almost all of them. In some cases, the distance to the pulsar is known from independent evidence such as parallax measurements, H I absorption, or because the distance is associated with an object of known distance such as a supernova remnant. The DM in these cases provides an average value of the electron density along the path. $\boxed{\mathcal{DM} = \int n_e ds \qquad \mathcal{EM} = \int n_e^2 ds}$

The DM and other results (such as Hα) have led to a model of electron density n_e in the Galaxy. The best current model for the electron density n_e for the Galactic plane is known as NE2025.

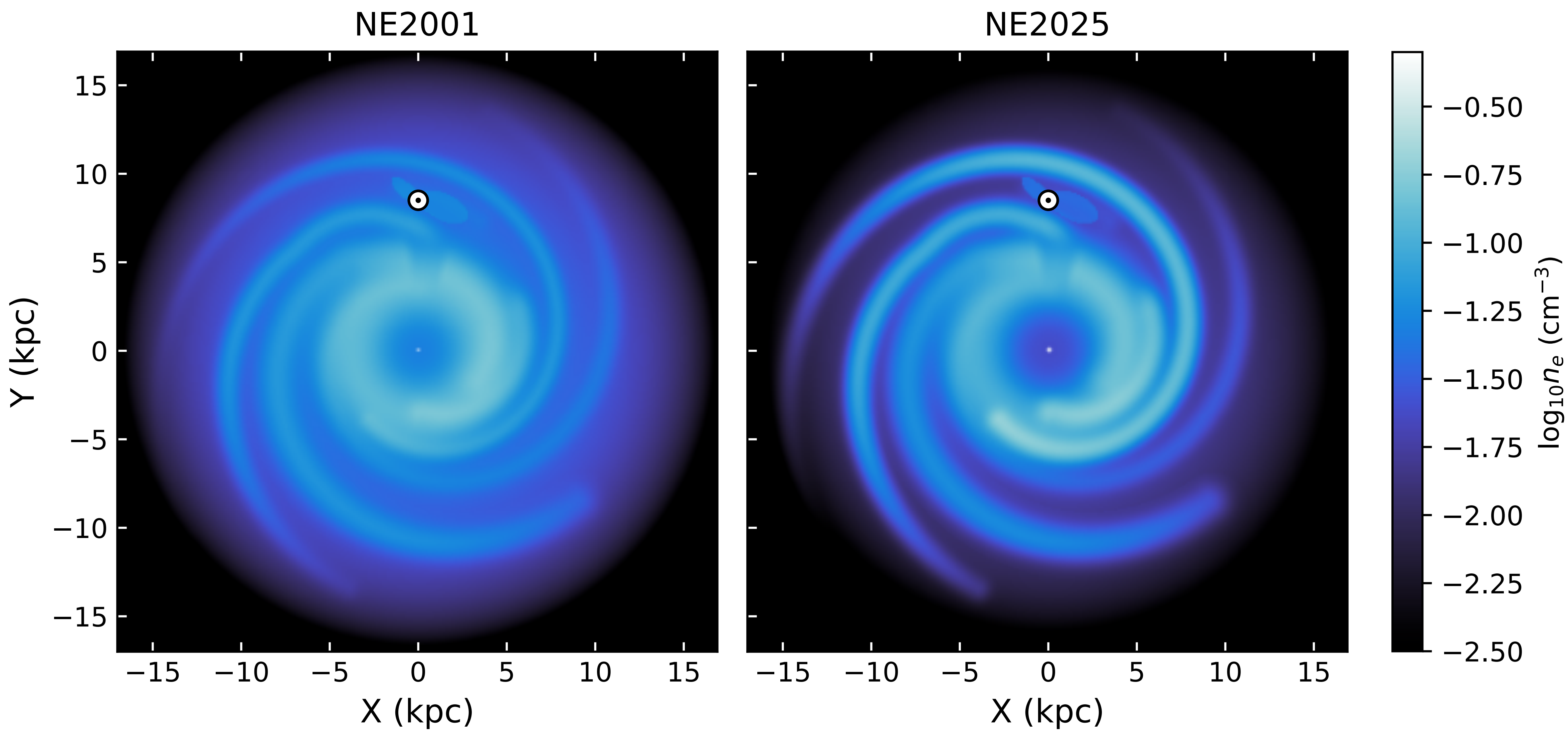


Contours of DM in units of pc cm^{-3} in the Galactic plane viewed from above the plane.

The sun and Galactic center (GC, cross) are marked; their separation is 8.5 pc.

Strong local enhancements of n_e, such as the GC at $\ell = 0$ or the Gum and Vela nebulae (SNR) at $\ell \approx 260°$, are reflected in the contours at distances beyond the enhancements. Similarly, voids such as those $\ell \approx 120°$ produce "fingers" in the contours at greater distance

Ocker & Cordes 2025, arXiv:2602.11838
NE2025 - https://github.com/stella-ocker/mwprop



Electron density in the Galactic plane (z = 0) for NE2001 (left) and NE2025 (right). The color stretch is the same for both panels.

Figure 1 in Ocker & Cordes (2026)

[References]

Taylor & Cordes (1993, ApJ, 411, 674)

Cordes & Lazio (2003, arXiv:astro-ph/0207156)

Schnitzeler (2012, MNRAS, 427, 664)

Ocker & Cordes (2026, ApJ, arXiv:2602.11838)

[Faraday Rotation - Propagation along a Magnetic Field]

- Now we extend the discussion of plasma propagation effects by considering the effect of an external, fixed magnetic field $\mathbf{B}_0$.

  The properties of the waves will then depend on the direction of propagation relative to the magnetic field direction. For this reason the plasma is called ***anisotropic***.

  Because of the magnetic field, a new frequency enters the problem, namely, the cyclotron frequency.

$$\text{cyclotron frequency} \qquad \omega_{\rm B} = \frac{eB_0}{m_e c} = 1.67 \times 10^7 \ (B_0/{\rm G}) \ {\rm s}^{-1}$$

$$\hbar\omega_{\rm B} = 1.16 \times 10^{-8} \ (B_0/{\rm G}) \ {\rm eV}$$

  If the fixed magnetic field $\mathbf{B}_0$ is much stronger than the field strengths of the propagating wave, then the equation of motion of an electron is approximately

$$m_e \frac{d\mathbf{v}}{dt} = -e\mathbf{E} - \frac{e}{c}\mathbf{v} \times \mathbf{B}_0$$

  For simplicity, assume that **the wave propagates along the fixed field $\mathbf{B}_0 = B_0\hat{\mathbf{z}}$**, and assume that **the wave is circularly polarized and sinusoidal**.

$$\mathbf{E}_{\pm}(t) = Ee^{-i\omega t}\left(\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}\right)$$

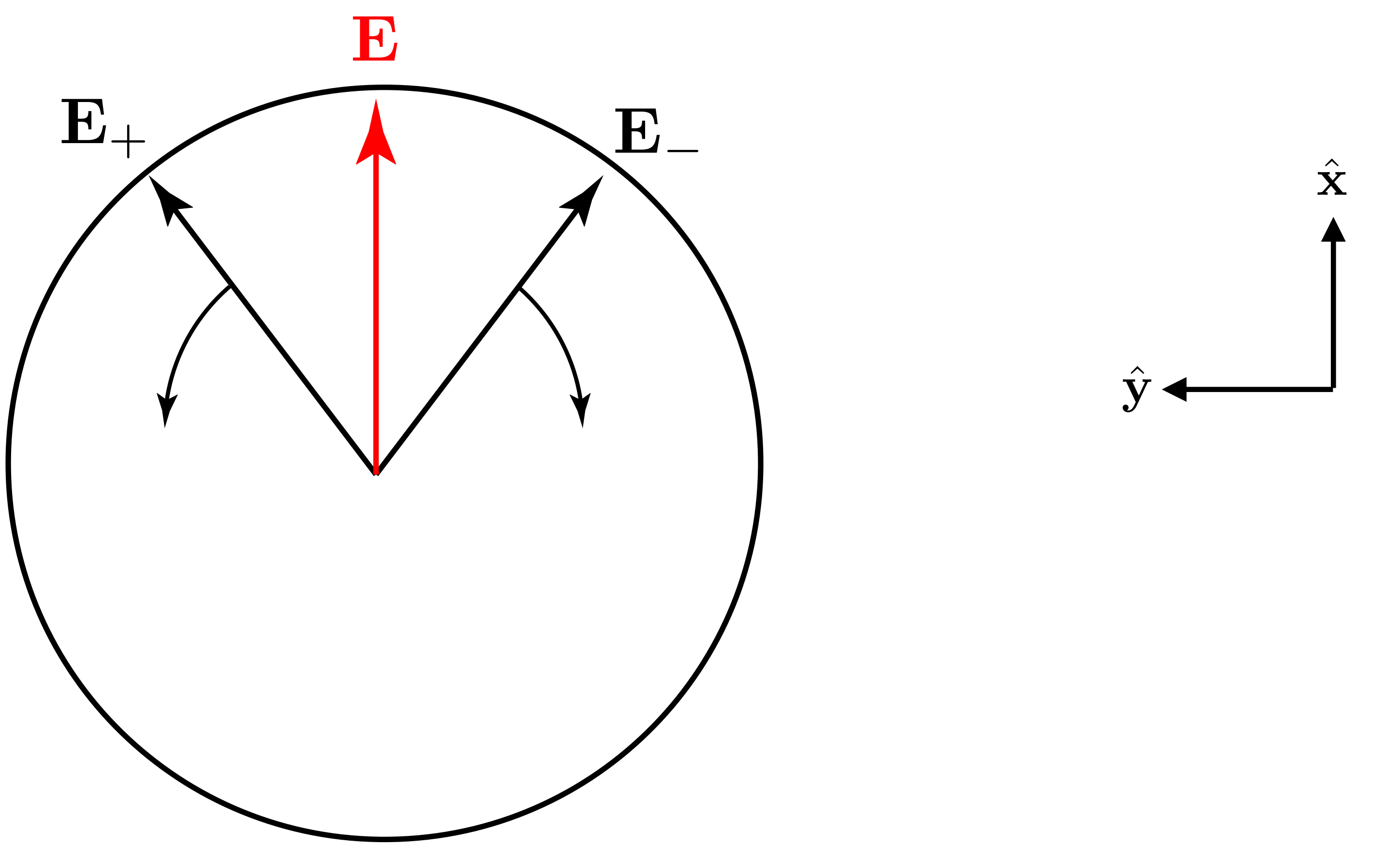
  Here, $\pm$ denotes right and left circular polarization.

  Note that Rybiki & Lightman define oppositely the RCP and LCP.

Bases vectors for circularly rotating electric field

The bases vectors $\mathbf{E}_{\pm}$ rotates counterclockwise (+) or clockwise (-) when viewed from the tip of the wave vector.

$$\begin{aligned} Re\left(\mathbf{E}_{\pm}\right) &= Re\left(\left(\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}\right) E e^{-i\omega t}\right) \\ &= Re\left(\left(\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}\right) E\left(\cos\omega t - i\sin\omega t\right)\right) \\ &\propto (\cos\omega t)\hat{\mathbf{x}} \pm (\sin\omega t)\hat{\mathbf{y}} \\ &\propto \cos(\pm\omega t)\hat{\mathbf{x}} + \sin(\pm\omega t)\hat{\mathbf{y}} \end{aligned}$$



Now, we solve the equation of motion for an electron and obtain the dielectric constant.

$$m_e \frac{d\mathbf{v}}{dt} = -e\mathbf{E} - \frac{e}{c}\mathbf{v} \times \mathbf{B}_0$$

$$\mathbf{E} \perp \mathbf{B}_0 \longrightarrow \quad \frac{d\mathbf{v}_\parallel}{dt} = 0 \qquad \mathbf{v}_\parallel = \text{constant}$$

$$\mathbf{v}_\perp = v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}}$$

$$(-i\omega)e^{-i\omega t}(v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}}) = -\frac{e}{m_e} E e^{-i\omega t}(\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}) - \frac{e}{m_e c} e^{-i\omega t}(v_y B_0 \hat{\mathbf{x}} - v_x B_0 \hat{\mathbf{y}})$$

$$\longrightarrow \quad v_x = -\frac{ie}{m_e \omega} E - \frac{ieB_0}{m_e c\omega} v_y, \quad v_y = \pm \frac{e}{m_e \omega} E + \frac{ieB_0}{m_e c \omega} v_x$$

$$(1)\ v_x = -\frac{ie}{m_e \omega} E - \frac{ieB_0}{m_e c\omega}\left(\pm\frac{e}{m_e\omega}E + \frac{ieB_0}{m_e c\omega} v_x\right)$$

$$\to \left(1 - \frac{e^2 B_0^2}{m_e^2 c^2 \omega^2}\right) v_x = -\frac{ie}{m_e \omega} E \left(1 \pm \frac{eB_0}{m_e c\omega}\right)$$

$$\to \left(\omega^2 - \omega_{\mathrm{B}}^2\right) v_x = -\frac{ie}{m_e} E \left(\omega \pm \omega_{\mathrm{B}}\right)$$

$$\to v_x = -\frac{ie}{m_e} E \frac{1}{\omega \mp \omega_{\mathrm{B}}}$$

$$(2)\ v_y = \pm\frac{e}{m_e \omega} E + \frac{ieB_0}{m_e c\omega}\left(-\frac{ie}{m_e\omega}E - \frac{ieB_0}{m_e c\omega} v_y\right)$$

$$\to \left(1 - \frac{e^2 B_0^2}{m_e^2 c^2 \omega^2}\right) v_y = \pm\frac{e}{m_e \omega} E \left(1 \pm \frac{eB_0}{m_e c\omega}\right)$$

$$\to \left(\omega^2 - \omega_{\mathrm{B}}^2\right) v_y = \pm\frac{e}{m_e} E \left(\omega \pm \omega_{\mathrm{B}}\right)$$

$$\to v_y = \pm\frac{e}{m_e} E \frac{1}{\omega \mp \omega_{\mathrm{B}}}$$

Finally, the current density and conductivity are obtained to be

$$\boxed{\therefore\ \mathbf{v} = \frac{-ie}{m_e\left(\omega \mp \omega_{\mathrm{B}}\right)} \mathbf{E}(t)}$$

$$\mathbf{j} \equiv -ne\mathbf{v} = \sigma \mathbf{E}$$

where $\sigma_{\mathrm{R,L}} \equiv \dfrac{ine^2}{m_e\left(\omega \mp \omega_{\mathrm{B}}\right)}$

Dielectric constant

$$\epsilon \equiv 1 - \frac{4\pi\sigma}{i\omega} = 1 - \frac{4\pi n e^2}{m_e\omega\left(\omega \mp \omega_{\rm B}\right)}$$

$$\epsilon_{\rm R,L} = 1 - \frac{\omega_p^2}{\omega\left(\omega \mp \omega_{\rm B}\right)}$$

Dispersion relation

$$\omega^2 = k^2c^2 + \frac{\omega_p^2\omega}{\omega \mp \omega_{\rm B}} \qquad \Leftarrow \quad c^2k^2 = \epsilon\omega^2 = \omega^2 - \frac{\omega_p^2\omega}{\omega \mp \omega_{\rm B}}$$

Phase velocity

$$v_p \equiv \frac{\omega}{k} = \frac{c}{n_r}$$

$$n_r = \sqrt{1 - \frac{\omega_p^2}{\omega\left(\omega \mp \omega_{\rm B}\right)}}$$

Right (+) and left (-) circularly polarized waves travel with different speeds.

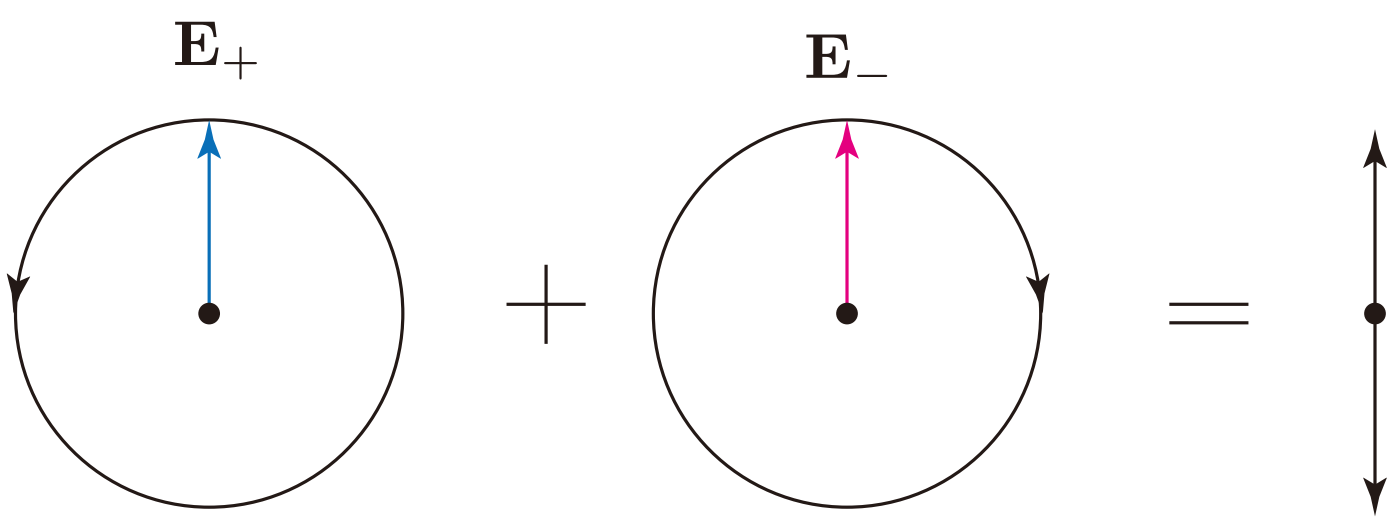
Speed difference sense is $v_{\rm R} > v_{\rm L}$.

These waves travel with different velocities. Therefore, a plane polarized wave, which is a linear superposition of a right-hand and a left-hand polarized wave, will not keep a constant plane of polarization, but **this plane will rotate as it propagates**. This effect is called **Faraday rotation**.

- **<u>Faraday Rotation</u>**

If the incident radiation is ***circularly polarized*** (either R or L), then the radiation will encounter different dispersion than unmagnetized case. But, the radiation will still remain circularly polarized.

If the incident radiation is ***linearly polarized***, i.e., a linear superposition of a right-hand and a left-hand polarized wave, then ***<u>the line of polarization will rotate as it propagates</u>***. This effect is called Faraday rotation.



$$\begin{aligned}\mathbf{E}(t) &= E e^{-i\omega t}\hat{\mathbf{x}} \\ &= \frac{1}{2}\left[(\hat{\mathbf{x}} + i\hat{\mathbf{y}}) + (\hat{\mathbf{x}} - i\hat{\mathbf{y}})\right] E e^{-i\omega t} \\ &= \frac{1}{2}\left(\boldsymbol{\varepsilon}_+ + \boldsymbol{\varepsilon}_-\right) E e^{-i\omega t}\end{aligned}$$

Decomposition of linear polarization into components of right and left circular polarization

- The phase angle ϕ after traveling a distance d is, in general, if the wave number is not constant along the path, is given by

$$\phi_{\mathrm{R,L}} = \int_0^d k_{\mathrm{R,L}} ds$$

Assume that $\omega \gg \omega_p$ and $\omega \gg \omega_{\mathrm{B}}$

$$k_{\mathrm{R,L}} = \frac{\omega}{c}\sqrt{\epsilon_{\mathrm{R,L}}} = \frac{\omega}{c}\left[1 - \frac{\omega_p^2}{\omega^2\left(1 \mp \omega_{\mathrm{B}}/\omega\right)}\right]^{1/2} \approx \frac{\omega}{c}\left[1 - \frac{\omega_p^2}{2\omega^2}\left(1 \pm \frac{\omega_{\mathrm{B}}}{\omega}\right)\right]$$
$$= \frac{\omega}{c} - \frac{\omega_p^2}{2c\omega} \mp \frac{\omega_p^2\omega_{\mathrm{B}}}{2c\omega^2} = k_0 \mp \frac{1}{2}\Delta k \quad \left(\Delta k \equiv k_{\mathrm{L}} - k_{\mathrm{R}} = \frac{\omega_p^2\omega_{\mathrm{B}}}{c\omega^2}\right)$$

Consider an electromagnetic wave that starts off linearly polarized in the x-direction at the source.

$$\mathbf{E}(t) = Ee^{-i\omega t}\hat{\mathbf{x}} = \frac{1}{2}\left[(\hat{\mathbf{x}} + i\hat{\mathbf{y}}) + (\hat{\mathbf{x}} - i\hat{\mathbf{y}})\right]Ee^{-i\omega t}$$

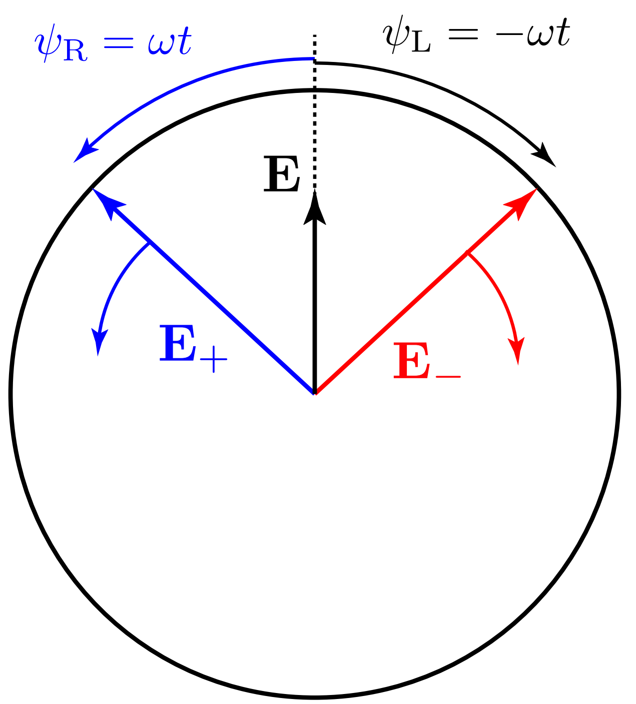
Let's define

$$\int_0^d k_{\mathrm{R,L}} ds = \int_0^d k_0 ds \mp \frac{1}{2}\int_0^d \Delta k ds \equiv \phi \mp \varphi$$

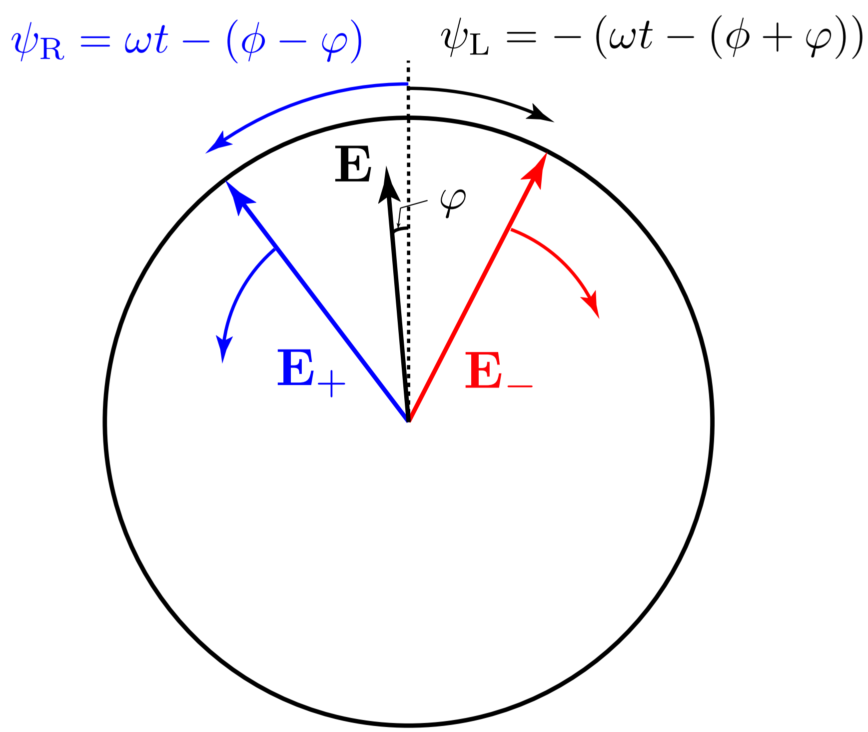
Then, after propagating a distance d through a magnetized plasma toward the observer, the electric field will be

$$\mathbf{E}(t) = \frac{1}{2}\left[(\hat{\mathbf{x}} + i\hat{\mathbf{y}})\,\overset{e^{i\int k_{\mathrm{R}} ds}}{e^{i(\phi-\varphi)}} + (\hat{\mathbf{x}} - i\hat{\mathbf{y}})\,\overset{e^{i\int k_{\mathrm{L}} ds}}{e^{i(\phi+\varphi)}}\right]Ee^{-i\omega t}$$
$$= (\hat{\mathbf{x}}\cos\varphi + \hat{\mathbf{y}}\sin\varphi)\,Ee^{i(\phi-\omega t)}$$

- The resulting electric field is also linearly polarized, but the polarization angle is rotated counterclockwise by angle φ (when viewed at a fixed position).



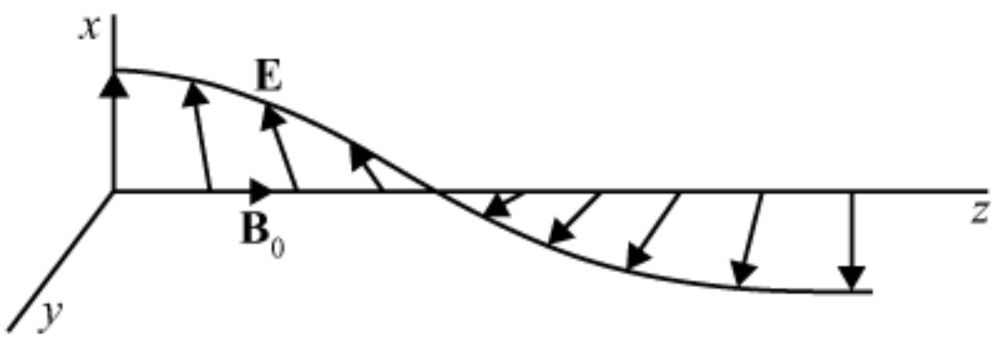
Before passing through the medium



After passing through the medium

- Radiation that starts linearly polarized in a certain direction is rotated by the *Faraday effect* through an angle φ after propagating a distance *d* through a magnetized plasma.

$$\boxed{\varphi = \frac{1}{2}\int_0^d \Delta k ds = \frac{1}{2}\int_0^d \frac{\omega_p^2 \omega_B}{c\omega^2} ds = \frac{2\pi e^3}{m_e^2 c^2 \omega^2}\int_0^d n B_\parallel ds}$$



- We cannot, of course, generally measure the absolute rotation angle, since we do not know the intrinsic polarization direction of the radiation when it started from the source.

However, since φ varies with frequency (as ω^{-2}), we can determine the value of integral $\int nB_{\parallel}ds$ by making measurements at several frequencies. This can give information about the interstellar magnetic field.

Rotation measure is defined by

$$\boxed{\varphi = \frac{2\pi e^3}{m_e^2c^2\omega^2}\mathcal{RM} = \frac{e^3\lambda^2}{2\pi m_e^2c^4}\mathcal{RM}, \text{ where } \mathcal{RM} \equiv \int_0^d n_eB_{\parallel}ds}$$

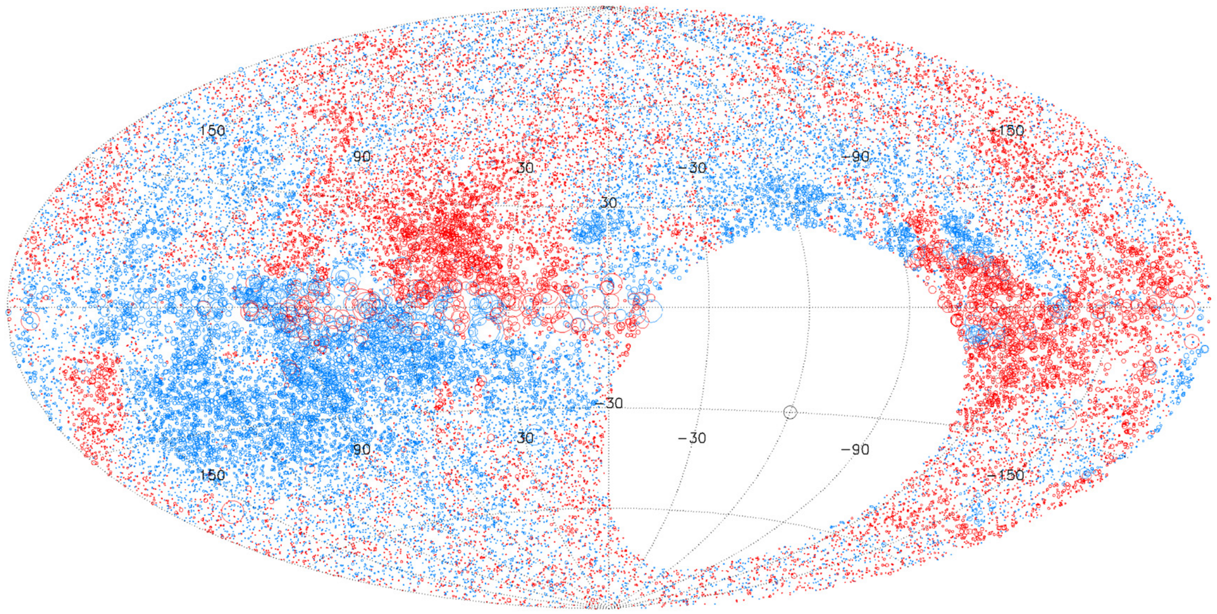
However, ***the field changes direction often along the line of sight and this method gives only a lower limit to actual field magnitudes.***

For measurements toward sources (pulsars) where the dispersion measure (DM) is also known, we can derive an estimate of the mean field strength along the line of sight.

$$\langle B_{\parallel}\rangle = \frac{\mathcal{RM}}{\mathcal{DM}} \quad \text{density-weighted magnetic field}$$

Radio astronomers have concluded that

$$\langle n_e\rangle \approx 0.03 \text{ cm}^{-3}$$
$$\langle B_{\parallel}\rangle \approx 3\mu\text{G}$$



(Taylor, Stil, & Sunstrum 2009, ApJ, 702, 1230)
Red circles are positive RM and blue circles are negative.
The size of the circle scales linearly with magnitude of RM.

Monte Carlo Methods for Radiative Transfer

Monte Carlo Methods

- **Monte Carlo Methods for Radiation Transport [Oleg N. Vassiliev]**

  https://download.e-bookshelf.de/download/0007/9138/48/L-G-0007913848-0016140947.pdf

- The Monte Carlo method was the name suggested by Metropolis in the late 1940th for a statistical approach to solving neutron diffusion and mutiplication problems that was being developed at that time at the Los Alamos Laboratory (Metropolis 1987).

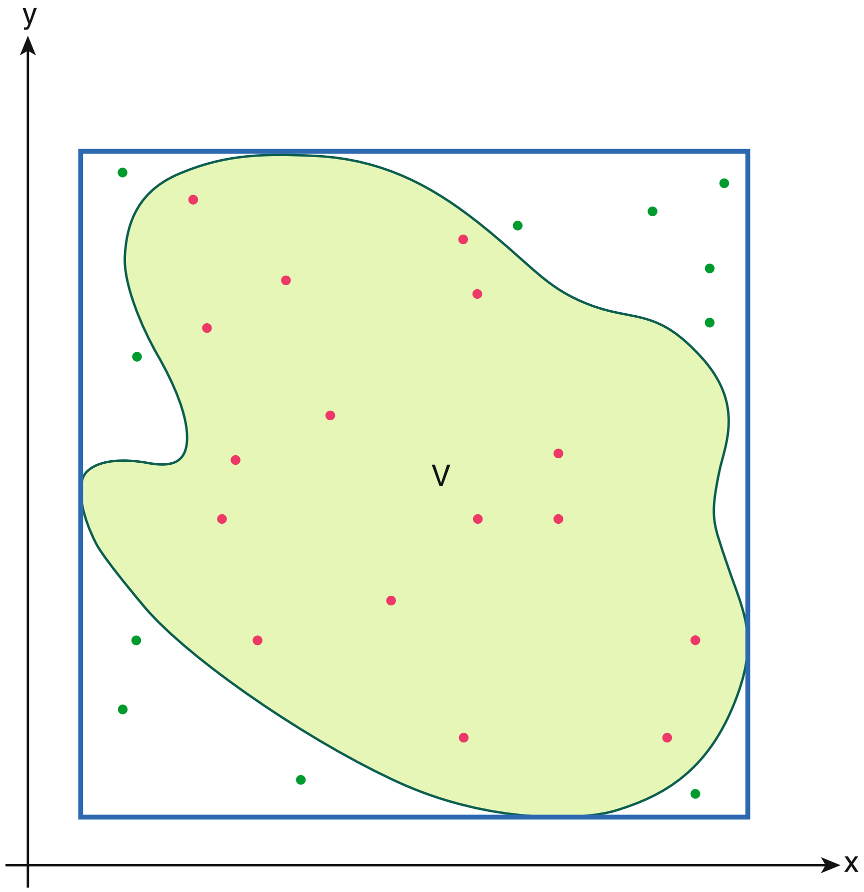
  - The method relies on statistical techniques such as sampling and inference. It has been used predominantly to study random phenomena. However, it has been originally proposed for studying differential equations, or more generally, integro-differential equations that arise in various branches of the natural sciences.

Example: Calculation of Volumes

Problem

Calculate volume V, Fig. 1.2.

Fig. 1.2 Calculation of volumes. The bounding volume V_B is the rectangle



Statistical Interpretation of the Problem

If a random point has a uniform distribution in a volume V_B (bounding volume) such that $V \subseteq V_B$, (the entire volume V is inside V_B), then the probability p of this point falling within V is equal to the ratio of the volumes, $p = V/V_B$. If N is the total number of random points uniformly distributed in V_B, and N_V is the number of points that fall within V, then the distribution of N_V is the binomial distribution with parameter p. Hence, again, the quantity of interest is expressed as function of parameter p of the binomial distribution, $V = pV_B$.

Sampling of the Distribution

To sample a random number N_V, we generate N random points with a uniform distribution in V_B and count the number of points N_V that fall within V. To simplify the sampling of a uniform distribution, a rectangular V_B is used in two dimensions, and an orthotope ("box") is used for higher dimensionalities. Sampling techniques are discussed in detail in the next chapter.

Estimator

For a sufficiently large N, we have $p \approx N_V/N$. Thus, the estimator of V is $\hat{\Theta}_V = V_B N_V/N$. This time the estimate is unbiased, i.e., $\mathrm{E}\{\hat{\Theta}_V\} = V$, where $\mathrm{E}\{.\}$ denotes the expectation.

Optimization

The optimization parameter for this problem is V_B. We, however, prefer to work with a dimensionless parameter, $x = V/V_B$, $0 \le x \le 1$. We need to find the variance of

the estimate as a function of x, and then find a value of x that minimizes the variance. Here we can use the formula for the variance of a binomial distribution

$$\boxed{\begin{aligned} E[k] &= np \\ \mathrm{Var}[k] &= npq = np(1-p) \end{aligned}}$$

$$\mathrm{Var}\left\{\frac{V_B}{N}N_V\right\} = \frac{V_B^2}{N^2}\mathrm{Var}\left\{N_V\right\} = \frac{V^2}{N}\left(\frac{1}{x}-1\right). \tag{1.4}$$

Thus, to minimize the variance we need to maximize x, by choosing the smallest V_B. At this point we can also find the fractional uncertainty

$$\frac{1}{V}\sqrt{\frac{V^2}{N}\left(\frac{1}{x}-1\right)} = \frac{1}{\sqrt{N}}\sqrt{\frac{1}{x}-1}. \tag{1.5}$$

The inverse square root of the sample size N is very common for fractional uncertainties in Monte Carlo calculations.

Probabilities (Appendix A)

A.1 Probability

Event A is the outcome of an experiment (trial). Sample space S is a set of all the possible outcomes of the experiment.

Example. If the trial is one roll of a die, then the sample space is $\{1, 2, 3, 4, 5, 6\}$.

The definition of the probability $P(A)$ of event A:

$$P(A) = \lim_{N \to \infty} \frac{n(A)}{N}, \tag{A.1}$$

where N is the number of trials and $n(A)$ is the number of times event A has occurred. The properties of probability:

1. $0 \leq P(A) \leq 1$.
2. $P(S) = 1$.
3. Additivity: if events A and B are mutually exclusive, that is $A \bigcap B = \emptyset$ (empty set), then $P(A \bigcup B) = P(A) + P(B)$.

A.2 Convergence in Probability

Definition (Dudewicz and Mishra 1988)**.** We say Z_n converges to Z in probability ($Z_n \xrightarrow{p} Z$) if for every $\epsilon > 0$

$$\lim_{n\to\infty} P\left\{|Z_n - Z| > \epsilon\right\} = 0.$$

A.3 Conditional Probability

The probability P of event A, given that event B has occurred:

$$P(A|B) = \frac{P\left(A \bigcap B\right)}{P(B)}. \tag{A.2}$$

A.4 Independent Events

Two events A and B are independent if $P(A|B) = P(A)$. If A and B are independent, then $P\left(A \bigcap B\right) = P(A) \cdot P(B)$.

A.5 The Total Probability Theorem

Let $\{A_i\}$, $i = 1, 2, \ldots, n$, be a complete set of mutually exclusive events. This means that $P\left(\bigcup_{i=1}^{n} A_i\right) = 1$ and $A_i \bigcap A_j = \emptyset$ for all $i \neq j$. Thus, in a trial one event from the set must occur, but any two events cannot occur. Then, for any event B:

$$P(B) = \sum_{i=1}^{n} P(A_i)\, P(B|A_i)\,. \tag{A.3}$$

A.6 Probability Distribution of a Discrete Variable

If the sample space is countable, that is all possible outcomes can be enumerated: $A_1, A_2, \ldots,$ then the probability distribution can be defined:

$$P(A_i) \equiv P(i) \equiv P_i\,; \quad i = 1, 2, \ldots \tag{A.4}$$

A.7 Probability Distribution Function

If the outcome of a trial is a number ξ, then the probability function $F_\xi(x)$ can be defined:

$$F_\xi(x) \equiv P\{\xi \leq x\}. \tag{A.5}$$

Random variable ξ may take a continuous range of values, a discrete set of values, or a combination of both. Variable x is continuous. Subscript ξ is often omitted.

The properties of the probability distribution function:

1.

$$F(x_2) \geq F(x_1)\,, \quad \text{if } \; x_2 > x_1; \tag{A.6}$$

2.

$$\lim_{x \to \infty} F(x) = 1; \tag{A.7}$$

3.

$$F(x) \geq 0, \quad \text{for } \forall\, x. \tag{A.8}$$

A.8 Probability Density Function

If the derivative of the probability distribution function $F(x)$ exists, then the probability density function $f(x)$ can be defined:

$$f_{\xi}(x) \equiv \frac{\mathrm{d}F_{\xi}}{\mathrm{d}x}. \tag{A.9}$$

It defines the probability of observing the random variable within a given interval:

$$P\{\xi \in \mathrm{d}x\} = f_{\xi}(x)\,\mathrm{d}x. \tag{A.10}$$

$$P(a \leq \xi \leq b) = \int_a^b f_{\xi}(x)\,\mathrm{d}x. \tag{A.11}$$

A.9 Transformation of Random Variables and Their Distributions

One-Dimensional Random Variables Given: random variable ξ has a probability density $f_\xi(x)$; random variable η (the transformed variable) is a known function of ξ, $\eta = g(\xi)$. Problem: find the probability density of η. Solution:

$$f_\eta(y) = f_\xi(x)\left|\frac{\mathrm{d}x}{\mathrm{d}y}\right| = f_\xi(x)\left|\frac{\mathrm{d}g}{\mathrm{d}x}\right|^{-1}. \tag{A.12}$$

The result should be written using y rather than x. This can be done through the substitution $x = g^{-1}(y)$.

Example.

$$f_\xi(x) = 2x\exp\left(-x^2\right), \quad x \ge 0. \tag{A.13}$$

Find the distribution $f_\eta(y)$, where $\eta = \xi^2$. Solution

$$f_\eta(y) = 2x\exp\left(-x^2\right)|2x|^{-1} = \exp(-y). \tag{A.14}$$

Two-Dimensional Random Variables Given: two random variables (ξ_1, ξ_2) that have a joint probability density $f_{\xi_1,\xi_2}(x_1, x_2)$; two random variables (η_1, η_2) (the transformed variables) are known functions of (ξ_1, ξ_2), $\eta_1 = g_1(\xi_1, \xi_2)$ and $\eta_2 = g_2(\xi_1, \xi_2)$. Problem: find the joint probability density of (η_1, η_2). Solution:

$$f_{\eta_1,\eta_2}(y_1, y_2) = f_{\xi_1,\xi_2}(x_1, x_2)\, |J|\,, \tag{A.15}$$

where J is the Jacobian:

$$J = \begin{vmatrix} \partial x_1/\partial y_1 & \partial x_1/\partial y_2 \\ \partial x_2/\partial y_1 & \partial x_2/\partial y_2 \end{vmatrix}. \tag{A.16}$$

To find the derivatives, we need to find the inverse transformation $\xi_1 = g_1^{-1}(\eta_1, \eta_2)$ and $\xi_2 = g_2^{-1}(\eta_1, \eta_2)$. Then

$$\frac{\partial x_i}{\partial y_j} = \frac{\partial g_i^{-1}}{\partial y_j}; \;\; i, j = 1, 2. \tag{A.17}$$

The result should be written using (y_1, y_2) rather than (x_1, x_2). This can be done through the substitutions $x_1 = g_1^{-1}(y_1, y_2)$ and $x_2 = g_2^{-1}(y_1, y_2)$.

Example. Transformation from Cartesian coordinates (x, y) to polar coordinates (ρ, ϕ). The inverse transformation: $x = \rho \cos\phi$, $y = \rho \sin\phi$. First, we find the Jacobian

$$J = \begin{vmatrix} \partial x/\partial\rho & \partial x/\partial\phi \\ \partial y/\partial\rho & \partial y/\partial\phi \end{vmatrix} = \rho. \tag{A.18}$$

Then, we can find the distribution:

$$f(\rho, \phi) = f(x(\rho, \phi), y(\rho, \phi))\, \rho. \tag{A.19}$$

For example, if $f(x, y)$ is a uniform distribution within a circle of radius R, that is

$$f(x, y) = \frac{1}{\pi R^2}, \text{ if } x^2 + y^2 < R^2. \tag{A.20}$$

then,

$$f(\rho, \phi) = \frac{\rho}{\pi R^2}, \text{ if } \rho < R. \tag{A.21}$$

A.10 Distribution of a Sum

If ξ and η are statistically independent random variables with distributions $f_\xi(x)$ and $f_\eta(x)$, then the distribution of the sum $\zeta = \xi + \eta$ is given by

$$f_\zeta(x) = \int_{-\infty}^{\infty} f_\xi\left(x'\right) f_\eta\left(x - x'\right) \mathrm{d}x' \,. \tag{A.28}$$

The integral above is called the convolution and is denoted as $(f_\xi * f_\eta)(x)$.

A.11 The Expectation Value

For a continuous random variable ξ the definition is:

$$\mathrm{E}\{\xi\} \equiv \int_{-\infty}^{+\infty} x f(x) \,\mathrm{d}x; \tag{A.29}$$

For a discrete variable:

$$\mathrm{E}\{\xi\} \equiv \sum_k k P_k. \tag{A.30}$$

The expectation value is also referred to as the mean value.

The properties of the expectation value:

$$\mathrm{E}\{\xi_1 + \xi_2\} = \mathrm{E}\{\xi_1\} + \mathrm{E}\{\xi_2\}. \tag{A.31}$$

$$\mathrm{E}\{C\xi\} = C\mathrm{E}\{\xi\}; \quad \text{where } C \text{ is a constant.} \tag{A.32}$$

A.13 The Normal Distribution

$$f_{\xi}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right]. \tag{A.40}$$

$$\mathrm{E}(\xi) = \mu; \quad \mathrm{Var}(\xi) = \sigma^2. \tag{A.41}$$

$$P\{\mu - \sigma < \xi < \mu + \sigma\} \approx 0.683. \tag{A.42}$$

$$P\{\mu - 2\sigma < \xi < \mu + 2\sigma\} \approx 0.954. \tag{A.43}$$

A.14 The Central Limit Theorem

If $\xi_1, \xi_2, \ldots, \xi_N$ are independent random numbers and each has an arbitrary distribution with mean μ and finite variance σ^2, and

$$\eta = \frac{1}{N} \sum_{i=1}^{N} \xi_i, \tag{A.44}$$

then in the limit $N \to \infty$, the distribution $f_{\eta}(x)$ tends to a normal distribution with $\mu_{\eta} = \mu$ and $\sigma_{\eta} = \sigma/\sqrt{N}$.

A.15 The Binomial Distribution

Bernoulli trials:

1. Trials are independent events.
2. Each trial results in exactly one of the two mutually exclusive outcomes, called success and failure.
3. The probability of success is constant from trial to trial.

The Bernoulli process, or the Bernoulli sequence, is a sequence of Bernoulli trials. The binomial distribution is the distribution of number of successes k in n Bernoulli trials:

$$P(k|n) = \binom{n}{k} p^k \, (1-p)^{n-k} \, ; \tag{A.45}$$

where $0 \leq k \leq n$, p is the probability of "success" in one trial, and

$$\binom{n}{k} = \frac{n!}{k! \, (n-k)!} \tag{A.46}$$

is the binomial coefficient.

expectation value:	$E[k] = np$
variance:	$\mathrm{Var}[k] = npq = np(1-p)$

A.16 The Poisson Distribution

It is derived from the binomial distribution by taking the limit $n \to \infty$, keeping the average number of successes, np, constant.

$$\boxed{\Leftarrow \mu = np}$$

$$P_k = \frac{\mu^k}{k!}\,\mathrm{e}^{-\mu}; \quad k = 0, 1, 2 \ldots \tag{A.47}$$

where μ is a parameter of the distribution, $\mu = \mathrm{E}\,(k) = \mathrm{Var}\,(k)$.

Example. The probability of k radioactive decays in 1 s is given by Eq. (A.47). In this case μ is the average number of decays per second (= activity in Bq).

The Poisson distribution is a discrete probability distribution that models the number of times a random, independent event occurs within a specific interval of time (distance, area, or volume).

Here, μ is the average rate of occurrence, while k is the actual number of successes.

- Derivation from the Binomial Distribution

  The Poisson distribution is obtained as a limit for binomial experiments with a very large number of trials (n) and a very low probability of success (p) per trial.

  - The probability of k successes in n trials is:

$$P(k|n) = \binom{n}{k} p^k (1-p)^{n-k}$$

  - Assuming a fixed average rate of occurrences $\mu = np$, we can replace p with μ/n;

$$P(k|n) = \frac{n!}{k!(n-k)!} \left(\frac{\mu}{n}\right)^k \left(1 - \frac{\mu}{n}\right)^{n-k}$$

  - Rearrange the terms:

$$P(k|n) = \frac{\mu^k}{k!} \left[\frac{n(n-1)\cdots(n-k+1)}{n^k}\right] \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-k}$$

  - Take the limit as $n \to \infty$:

$$\frac{n(n-1)\cdots(n-k+1)}{n^k} \to 1, \quad \left(1 - \frac{\mu}{n}\right)^n \to 1, \quad \left(1 - \frac{\mu}{n}\right)^{-k} \to e^{-\mu}$$

  - Resulting Poisson PDF:

$$\boxed{P(k|\mu) = \frac{\mu^k e^{-\mu}}{k!}}$$

Alternative derivation:
https://www.pp.rhul.ac.uk/~cowan/stat/notes/PoissonNote.pdf

A.17 The χ^2 Distribution

If $\xi_i \ (i = 1, 2, \ldots, r)$ have normal distributions with the mean equal to 0, and the variance equal to 1, then the sum

$$\chi^2 = \sum_{i=1}^{r} \xi_i^2 \tag{A.48}$$

is χ^2 distributed. Parameter of the distribution r is called the number of degrees of freedom.

$$f_{\chi^2}(x) = \frac{x^{r/2-1} \exp(-x/2)}{\Gamma(r/2)\, 2^{r/2}}. \tag{A.49}$$

$$\mathrm{E}\left(\chi^2\right) = r. \tag{A.50}$$

$$\mathrm{Var}\left(\chi^2\right) = 2r. \tag{A.51}$$

A.18 Other Characteristics of a Distribution

The most probable value. It corresponds to the maximum of the distribution. It is often, but not always, can be found by solving:

$$\frac{\partial f}{\partial x} = 0. \tag{A.52}$$

The median, $x_{1/2}$.

$$\frac{1}{2} = \int_{-\infty}^{x_{1/2}} f(x)\,\mathrm{d}x; \tag{A.53}$$

The lower, $x_{1/4}$, and upper, $x_{3/4}$, quartiles.

$$\frac{1}{4} = \int_{-\infty}^{x_{1/4}} f(x)\,\mathrm{d}x; \quad \frac{3}{4} = \int_{-\infty}^{x_{3/4}} f(x)\,\mathrm{d}x. \tag{A.54}$$

The quantile, x_q.

$$q = \int_{-\infty}^{x_q} f(x)\,\mathrm{d}x; \quad 0 < q < 1. \tag{A.55}$$

The interquartile range (IQR) = $x_{3/4} - x_{1/4}$.

A.19 Statistics

The statistic is a quantity, which is calculated from observed data. In the following examples: $\{\xi_1, \xi_2, \ldots \xi_N\}$ is the observed data, or the sample, and N is the sample size.

The sample average.

$$\bar{\xi} = \frac{1}{N} \sum_{i=1}^{N} \xi_i. \tag{A.56}$$

The sample median.

$$\xi_{1/2} = \begin{cases} \xi_{\frac{N+1}{2}}, & \text{if } N \text{ is odd} \\ \frac{1}{2}\left(\xi_{\frac{N}{2}} + \xi_{\frac{N}{2}+1}\right), & \text{if } N \text{ is even.} \end{cases} \tag{A.57}$$

The sample in Eq. (A.57) is assumed to be sorted in the increasing order: $\xi_1 \leq \xi_2 \leq \cdots \leq \xi_N$.

Sample quartiles and quantiles. The sample median as well as sample quartiles and quantiles are examples of order statistics. Order statistics are defined by the location of a random value in a sorted sample.

The sample variance, unbiased.

$$s^2 = \frac{1}{N-1} \sum_{i=1}^{N} \left(\xi_i - \bar{\xi}\right)^2. \tag{A.58}$$

A.20 Point Estimators

Hypothesis: ξ has a distribution $f_\xi(x, a, b, \ldots)$, where $a, b, \ldots$ are unknown parameters.

Problem. Given: Sample $\{\xi_1, \xi_2 \ldots \xi_N\}$. Estimate $a, b, \ldots$.

For each parameter $a, b, \ldots$ an estimator that is a function of the sample $\hat{\Theta}_a(x_1, x_2, \ldots x_N)$, etc., should be designed. Good estimators are:

1. Unbiased, $\mathrm{E}\{\hat{\Theta}_a(\xi_1, \xi_2, \ldots \xi_N)\} = a$.
2. Efficient, $\mathrm{Var}\{\hat{\Theta}_a\}$ is minimal.
3. Robust. Unusually small or large values in the sample have limited influence on the result. Other characteristics of robustness also exist, for example, see Hampel (1986).

A.21 The Maximum Likelihood Method

It is a method for deriving estimators that is based on the assumption that the acquired sample is the most probable. For example, if the probability distribution has two parameters, $f(x, a, b)$, then:

1. The likelihood function is written:

$$L(a, b) = \prod_{i=1}^{N} f(\xi_i, a, b). \tag{A.59}$$

2. The derivatives are calculated and a system of two equations is written

$$\frac{\partial L}{\partial a} = 0; \quad \frac{\partial L}{\partial b} = 0. \tag{A.60}$$

3. The system is solved for a and b. Then the expression for a is the estimator for a, $\hat{\Theta}_a(\xi_1, \xi_2, \ldots \xi_N)$. Similarly, the expression for b is $\hat{\Theta}_b$.

The algebra is often simpler if $\ln L$ is used instead of L. Moreover, the derivative equal to 0 does not always give a maximum. Sometimes the derivative does not even exist. The main advantage of this approach is that the most likelihood estimators are asymptotically efficient.

Example. Estimate parameter μ of the normal distribution.

First, we write the likelihood function:

$$\ln L(\mu, \sigma) = \sum_{i=1}^{N} \ln f(\xi_i, \mu, \sigma^2), \tag{A.61}$$

where $f(\xi_i, \mu, \sigma^2)$ is a normal distribution with parameters μ, σ^2. Then we solve the equation

$$\frac{\partial \ln L}{\partial \mu} = 0 \tag{A.62}$$

for μ. The solution is:

$$\mu = \frac{1}{N} \sum_{i=1}^{N} \xi_i; \tag{A.63}$$

Then, estimator for μ is the sample average

$$\hat{\Theta}_\mu = \frac{1}{N} \sum_{i=1}^{N} \xi_i. \tag{A.64}$$